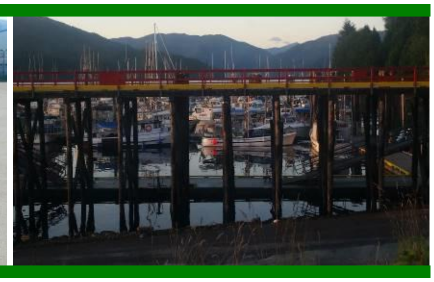
EA3642

North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2015

FINAL REPORT



Prepared for:

North Coast - Skeena First Nations Stewardship Society
Prince Rupert, BC

Prepared by:

David Robichaud,[1] Angela Addison[2] and Matthew Imrie[1]

[1] LGL Limited environmental research associates
9768 Second Street, Sidney, BC, V8L 3Y8

[2] North Coast - Skeena First Nations Stewardship Society
612, 2 Ave W, Prince Rupert, BC, V8J 1H2

2 March 2016

EXECUTIVE SUMMARY

In 2015, the North Coast Skeena First Nations Stewardship Society (NCSFNSS) operated a recreational fishery monitoring program in Pacific Fishery Management Areas 3 and 4. The program consisted of an access point creel survey which included concurrent ground and aerial surveys. Estimates were produced of charter and independent recreational fishery effort and harvest in-season for the months of June, July and August of 2015. Lodge based recreational harvest and effort was monitored by DFO via the collection of electronic daily fishing data submitted at the conclusion of the fishing season. Recreational lodge harvest and effort census data were integrated with creel survey data to produce harvest and effort estimate totals for all types of recreational fishing activity occurring within the study.

Data collection and estimation analytical methodologies are documented in this report. Survey design and implementation were consistent with those operated by DFO during the past seven fishing seasons. Analytical methods were similar to those used in prior years but not identical.

The 2015 Area 3 and 4 creel survey started on June 1st and ended on August 31st. 2,569 vessel interviews (involving 8,149 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 158 survey shifts. 24 aerial effort surveys were conducted with 60% of flights scheduled on weekdays and 40% on weekend days or holidays. Angler interviews focused on obtaining catch and effort data for Pacific salmon (*Oncorhynchus* spp.), Pacific Halibut (*Hippoglossus stenolepsis*), Lingcod (*Ophiodon elongates*) and Rockfish (*Sebastes* spp.). Interviews also provided opportunities to collect Chinook Salmon (*O. tshawytscha*) and Pacific Halibut biological samples and measurements. High priority was placed on gathering Chinook Salmon scale samples, length measurements, and checking for adipose fin clips. Pacific Halibut length measurements were also sought.

Total fishing effort was estimated to be 11,585 vessel trips. In total, an estimated 14,987 Chinook Salmon were caught, with 12,756 fish kept and 2,231 released. It was estimated that 35,685 Coho Salmon (*O. kisutch*) were caught, with 32,148 fish kept and 3,537 released. Total Halibut catch was estimated to be 17,322 individuals, with 12,848 fish kept and 4,475 released. There were also total retained catch estimates of 3,051 lingcod, 6,712 rockfish, and 492 'other groundfish'.

Creel surveyors collected scale samples from 913 Chinook Salmon, and length measurements from 903 individuals. Also, 995 Chinook Salmon were inspected for adipose fin clips, of which 133 were clipped. In addition, 23 Coho Salmon were inspected for adipose fin clips, of which 4 were clipped. Also, 730 Pacific Halibut were measured for length, and corresponding weights were estimated using the International Pacific Halibut Commission's length-weight conversion formula. The average net weight of sampled Pacific Halibut was calculated to be 14.4 pounds in Area 3, and 16.8 pounds in Area 4.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

INTRODUCTION

The recreational fishery accessed via Prince Rupert and Port Edward is popular among Canadian and international visitors, as well as local area anglers seeking opportunities to target and harvest salmon, groundfish and shellfish. This fishery occurs in DFO's Pacific Region, in Management Areas 3 and 4.

During each recreational fishing season from 2008 to 2014, Fisheries and Oceans Canada (Prince Rupert) has estimated fishing effort and species-specific catch rates, using creel survey methodologies (e.g., Van Tongeren and Winther 2010). Starting in 2015, the responsibility for carrying out the creel survey shifted to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey followed the same design used by Fisheries and Oceans Canada in the past, which was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002). LGL Limited was engaged to support the 2015 project by developing software utilities for data entry, compilation and analysis, in consultation with NCSFNSS biologists. NCSFNSS led collection of angler interview and aerial survey effort data, and was responsible for data review, data entry, in-season reporting of preliminary estimates to DFO, and submission of biological samples to DFO and to the CWT Salmon Sport Head Recovery Program.

This report documents the 2015 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

SCOPE

The geographic scope of the creel survey was the same as in the previous seven years (Van Tongeren and Winther 2010), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island; Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Ares 4 (4A-4H, and 4L).

The temporal scope of the full study was from June 1st through August 31st, 2015. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea.

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook (*Oncorhynchus tshawytscha*) and Coho salmon (*O. kisutch*), Pacific Halibut (*Hippoglossus stenolepsis*) and Lingcod (*Ophiodon elongates*).

GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used by DFO over the last seven years (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from June through August 2015. The precision goal for the study was to estimate total harvests within 25% of the true value 19 times out of 20. Creel survey strata included temporal separation by month and day type (weekdays vs. weekend days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

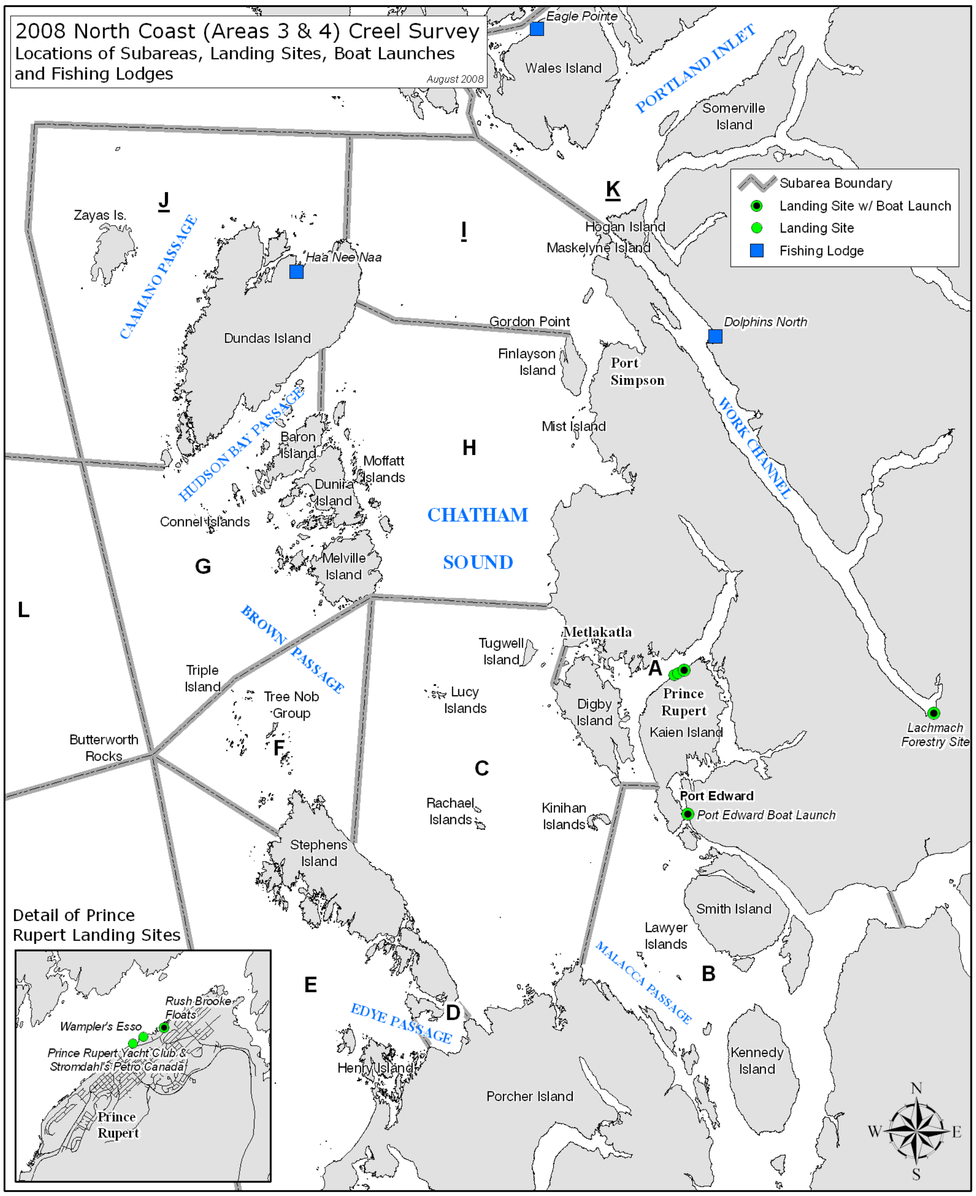


Figure 1. **Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas, which were used as spatial strata in the creel survey design.**

The eight objectives for this study were:

1) Develop software for data management and analysis comparable to DFO's CREST program;
2) Develop and implement standardized training protocols for creel survey procedures;
3) Collect biological data, samples and/or measurements from as many Chinook Salmon, Coho Salmon, or Pacific Halibut as possible during angler interviews;
4) Provide DFO with monthly in-season catch estimates for all salmon species and Pacific Halibut harvested in the North Coast recreational fishery;
5) Provide estimates of the retained and released catch of all species targeted by the recreational fishery and associated estimates of precision;
6) Provide communication materials to recreational harvesters to convey the use and importance of their catch data in fisheries management;
7) Produce a final written report to be available to a broad audience; and
8) Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests and First Nations.

Objectives 1 and 2 were required in order to meet any subsequent objective. Objective 3 was met throughout the study period, as fish were biosampled at access points. Scale samples, heads from adipose-clipped Chinook and Coho salmon, and other biological data were submitted to DFO as frequently as possible throughout the study period. Similarly, Objective 4 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 5 was met by completing this report. Objectives 6, 7 and 8 will be met subsequent to the publication of this report, as the contents of the former will be based on the results presented herein.

METHODS

A complete census of catch and effort was available for lodge-based anglers. However, the remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each month and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the number of lodge-based boats known to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1) conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;

- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea, and how many of each had adipose fins clipped;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, kept fish were measured and sexed. Chinook and Coho salmon were checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook Salmon. Chinook Salmon were also examined for flesh colour;
- Refusals – whether or not the anglers refused an interview, or allowed catch to be biosampled; and
- Weather conditions.

The methods used were adapted from those developed and documented for the Strait of Georgia Creel Survey (English et al. 2002). Statistical precision associated with creel survey catch and effort estimates were based on those documented in English et al. (2002) and Blakley et al. (2003).

This procedure provides a statistically unbiased estimate of catch per effort, provided the anglers interviewed are representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that was conducted from that site in recent years (see Van Tongeren and Winther 2010). Wherever possible, the busiest access points were selected preferentially in order to obtain the maximum number of interviews. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Ideally, interviews would capture data about angler activity in all creel subareas.

Interviewing sessions were to be separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00 and PM shifts from 13:00 to 21:00. In total, 148 shifts were scheduled.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 2-3 weekend days and usually on 5-6 weekdays each month. Timing of the overflights was targeted for peak fishing periods, but may have been off-peak on some occasions due to weather or scheduling conflicts.

Angler Activity Patterns

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002).

The first weighting factor, $W1$, expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site *a*, were consistent for landing site *a*, even during the days when no interviews occurred. A specific $W1$ was calculated for each landing site during each month and day type:

$$W1_{mda} = \frac{N_{md}}{K_{mda}} \qquad \text{(Eqn. 1)}$$

where N_{md} was the total number of type *d* days in month *m*; and K_{mda} was the number of days during which interviews occurred at landing site *a*, on type *d* days during month *m*.

The second weighting factor, $W2$, expanded the numbers of interviews conducted, to account for the boats that were *not* interviewed (or that refused interviews). That is, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific $W2$ was calculated for each surveyor (*u*) operating on each surveying date (*k*) at each landing site (*a*):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku}\left(\dfrac{LF_{aku}}{LF_{aku} + LN_{aku}}\right)}{LF_{aku} - ri_{aku}} \qquad \text{(Eqn. 2)}$$

,

where LF_{aku} was the number of interviewed boats that were fishing (a subset of these, ri_{aku}, refused an interview, or had an incomplete interview that could not be used), LN_{aku} was the number of interviewed boats that were not fishing, and LM_{aku} was the number of boats that were not interviewed by surveyor *u*, during surveying date *k*, at landing site *a*.

We used the term $A_{mdsakut}$ to denote the number of non-lodge boats reporting activity during time-block *t*, as interviewed by surveyor *u* on survey date *k*, at landing site *a*, in subarea *s*, during month *m*, and on day type *d* (n_{mdsaku} was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{mdst} = \sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot A_{mdsakut} \right) \qquad \text{(Eqn. 3)}$$

.

Summing the adjusted number of anglers over the 16 time-blocks gave:

$$T'_{mds} = \sum_t A'_{mdst} \,. \qquad \text{(Eqn. 4)}$$

The proportion of non-lodge boats (P_{mdst}) that were actively fishing during in each of 16 hourly time-blocks (*t*) was calculated for each month, day type, and subarea:

$$P_{mdst} = \frac{A'_{mdst}}{\sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku}\right)} \qquad \text{(Eqn. 5)}$$

Using this method, 72 unique angler activity patterns were to be estimated (i.e., 3 months × 2 day types × 12 subareas). To reliably describe angler activity, a relatively large number of anglers (~ 60) needed to be interviewed in each of the 72 blocks. In the end, some blocks contained too few interviews (Table 1), so it was decided to pool activity data over subarea. The equation for non-lodge angler activity was thus

$$P_{mdt} = \frac{\sum_s A'_{mdst}}{\sum_s \sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku}\right)}, \qquad \text{(Eqn. 6)}$$

and the associated variance was:

$$S^2_{P_{mdt}} = \frac{(P_{mdt})(1 - P_{mdt})}{\sum_s \sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku}\right)}. \qquad \text{(Eqn. 7)}$$

Catch Per Effort Estimation

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species of fish from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview (i), the guided status (g), month (m) and the catch (C) of each species (r) in each subarea (s) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{msrgi} = \frac{C_{msrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\}. \qquad \text{(Eqn. 8)}$$

Mean CPE was calculated as the average catch, over all n_{msg} boat trips:

$$C\hat{P}E_{msrg} = \frac{\sum_{i=1}^{n_{msg}} C_{msrgi}}{n_{msg}}. \qquad \text{(Eqn. 9)}$$

The variance for the estimate of mean catch per effort was calculated as:

$$S^2_{C\hat{P}E_{msrg}} = \frac{\sum \left(CPE_{msrgi} - C\hat{P}E_{msrg}\right)^2}{(n_{msg} - 1)}. \qquad \text{(Eqn. 10)}$$

Table 1. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns and CPE, for all levels of each factor.

		Subarea											
Month	*Day Type*	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
June	*WD*	13	18	3	28	39	34	15	1	93	11	12	6
	WE	12	10	4	26	60	44	25	8	81	9	15	9
July	*WD*	35	23	8	33	60	66	71	4	158	20	21	17
	WE	20	18	4	32	50	46	44	1	122	13	30	12
August	*WD*	17	15	5	25	31	40	24	6	123	12	24	17
	WE	9	13	11	22	41	53	40	3	130	7	27	14

In the event that a month/subarea-specific sample size was < 5, adjacent subareas would be pooled together as follows:

- if 3I, 3J or 3K are too low, then then it will be combined with the other two to give a 3IJK regional average CPE;
- if 4A, 4B or 4C are too low, then then it will be combined with the other two to give a 4ABC regional average CPE;
- if 4D, 4E or 4F are too low, then then it will be combined with the other two to give a 4DEF regional average CPE; and
- if 4G, 4H or 4L are too low, then then it will be combined with the other two to give a 4GHL regional average CPE.

Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip.

Angler Effort Estimation

A complete census of lodge-based angling effort were collected from lodge log books by DFO personnel, and are summarized in this report.

To obtain statistically valid estimates of non-lodge fishing effort, anglers were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, non-lodge boats needed to be excluded from the overflight counts. To do that, we needed to estimate how many of the lodge boats were visible by each of our overflights. Two days out of each week were lodge 'change-over' days. On these days, no lodge-based vessels were fishing during our overflights (departing clients left before our flight and new clients arrived after it). On all other days, lodge vessels could have been fishing during our flight. Lodges reported their mid-day activity (which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day), allowing us to estimate that ~60% of the vessels on 'non-change-over days' were probably fishing during our overflights.

Table 2 shows the number of overflight surveys scheduled for each month and day type. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of recreational and commercial vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey o (conducted during month m and on day type d), observers tallied the total number of recreational boats that were actively fishing at time t on their i^{th} pass over subarea s, V_{mdsoit}. For each visit to a subarea within Management Area 3, the relative proportion of boats was calculated, ρ_{mdsoit}, and the total number of lodge boats (λ_{mdo}) apportioned accordingly:

$$\lambda_{mdsoit} = \rho_{mdsoit} \cdot \lambda_{mdo} \quad . \qquad \text{(Eqn. 11)}$$

The number of non-lodge boats was hence:

$$v_{mdsoit} = V_{mdsoit} - \lambda_{mdsoit} \, . \qquad \text{(Eqn. 12)}$$

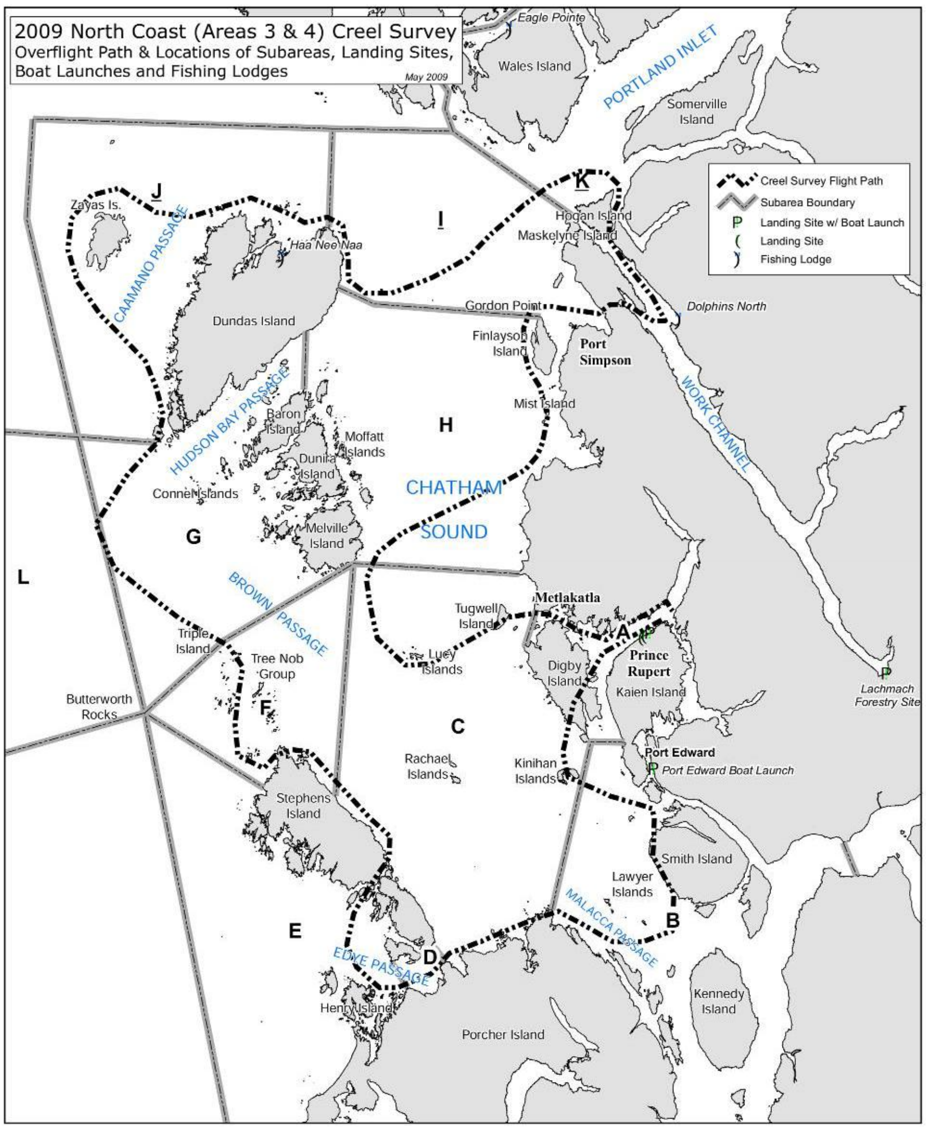


Figure 2. **Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert, and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).**

Table 2. Summary of overflight sampling effort (number of flights shifts by month and day type).

Month	Weekday	Weekend/ Holiday	Total
June	5	3	8
July	5	3	8
August	6	2	8
Total	16	8	24

Since angling occurs over the course of the entire day, the number of non-lodge boats that were observed was divided by the proportion of average daily number of boats active (P_{mdst}) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, hence the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea, B_{mdso}:

$$B_{mdso} = \sum_{i} \frac{v_{mdsoit}}{P_{mdst}} \quad . \qquad \text{(Eqn. 13)}$$

The variance of B_{mdso} was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned}&\text{if } z = x/y,\ Var(z) = (y^{-2})Var(x) + (x^2 y^{-4})Var(y)\ \ ;\\&\text{if } z = xy,\ Var(z) = (y^2)Var(x) + (x^2)Var(y)\ \ ;\ \text{and} \qquad \text{(Eqn. 14)}\\&\text{if } z = cx,\ Var(z) = (c^2)Var(x) \quad \text{where c is a constant.}\end{aligned}$$

Since there was no variance associated with the boat counts,

$$S^2_{B_{mdso}} = (v_{mdsoit})^2 (P_{mdst})^{-4} S^2_{P_{mdst}} \ . \qquad \text{(Eqn. 15)}$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys conducted, n_{mds}, as:

$$\hat{B}_{mds} = \frac{\sum_{o=1}^{n_{mds}} B_{mdso}}{n_{mds}} \ , \qquad \text{(Eqn. 16)}$$

with a variance of:

$$S^2_{\hat{B}_{mds}} = (n_{mds})^{-2} S^2_{B_{mdso}} \ . \qquad \text{(Eqn. 17)}$$

Total monthly fishing effort, was calculated for each day type and subarea by multiplying the average daily effort by the number days of day type *d* that occurred in month *m*:

$$E_{mds} = \hat{B}_{mds} \cdot N_{md} \ . \qquad \text{(Eqn. 18)}$$

with a variance of:

$$S^2_{E_{mds}} = S^2_{\hat{B}_{mds}} \cdot N^2_{md} \qquad \text{(Eqn. 19)}$$

The standard error of the estimate of the total monthly fishing effort, after pooling over day types, was:

$$S_{E_{ms}} = \sqrt{\frac{\sum_d S^2_{E_{mds}}}{\sum_d n_{mds}}} . \qquad \text{(Eqn. 20)}$$

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of month and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of month and day type during which the data gap occurred.

Catch and Harvest Estimation

Total catch was calculated for each month, subarea and species by multiplying total angling effort by catch per effort, and then summing over day type:

$$C_{msr} = \sum_d \left(E_{mds} \cdot C\hat{P}E_{msrg} \right) \qquad \text{(Eqn. 21)}$$

where g = 'all' (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{msr}} = \sqrt{\sum_d \left(E^2_{mds} \frac{S^2_{CPE_{msrg}}}{n_{msg}} + CPE^2_{msrg} \frac{S^2_{E_{mds}}}{n_{mds}} + \frac{S^2_{CPE_{msrg}}}{n_{msg}} \frac{S^2_{E_{mds}}}{n_{mds}} \right)} \qquad \text{(Eqn. 22)}$$

where n_{mds} and n_{msg} are the sample sizes for E and CPE estimates, respectively.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released.

Biosampling

When given permission by the anglers, observers collected biological data from Chinook and Coho salmon and Pacific Halibut. All three species were sexed. Chinook and Coho salmon were examined for clipped adipose fins (incidence data used by DFO for coded wire tag analyses) and head tags. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook Salmon (used by DFO to determine age, and stock composition).

Biomass of Harvested Pacific Halibut

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\text{Weight(lb)}_{\text{net}} = 6.921\times10^{-6} \times \text{FL(cm)}^{3.24}$$
$$\text{Weight(lb)}_{\text{round}} = 9.166\times10^{-6} \times \text{FL(cm)}^{3.24} \qquad \text{(Eqn. 23)}$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

Statistical Analyses

Comparisons of CPE, effort, catch or harvest by month or day type were performed within each fish taxa using Kruskal-Wallis tests (Zar 1984). When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

RESULTS

Angler Interviews

Over the three month study period, there were 158 observer shifts, during which 2,569 interviews were conducted, involving 8,149 non-lodge anglers (Table 3; Figure 3). July was the busiest month, with 1,058 interviews conducted (average 18.6 interviews per shift), which represented 41% of the total number of study period interviews (Figure 1). In all there were 58 refusals, or 2.3% refusal rate. The highest refusal rate (3.4%) was at Rushbrooke Boat Launch, where 66% of the refusals occurred. The refusal rates for Prince Rupert Yacht Club and Port Edward Boat Launch were 1.9% and 0.8%, respectively.

On average, 34% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 3 of 733 trips interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 41% and 56% of the time, respectively (Table 4).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F being the overwhelming favourite (Table 5). Subareas 4B, 4C, and 4D were the next most likely areas for anglers to have reported activity. The subarea with the lowest number of trips reporting activity was 4E. Only 46 interviews were of parties that only targeted shellfish, and those were largely in subarea 4A (Table 6).

Angler Activity Patterns

Monthly non-lodge angler activity patterns estimates are shown in Figure 4. There was virtually no difference between the weekday and weekend/holiday activity patterns for June, July and August. Angler activity patterns showed some changes across the three-month survey period. As expected, there was a shift over time away from late-afternoon angling with the reduction in daylight. In August, activity patterns were flat, with no distinct peak between 8 AM and 2 PM.

Catch Per Effort Estimates

In all months and all management areas, sample sizes were adequate (n >=5) to produce reliable estimates of non-lodge CPE (Table 7). However, when the interview data were split between guided and non-guided groups, monthly sample sizes did not suffice in some of the management areas (e.g., 3K, 4B-E) where guiding was less common (Table 7). In those cases, data from three adjacent subareas were pooled together for CPE calculations (Table 8). Average number of fish

Table 3. Monthly number of shifts numbers of boats interviewed per landing site, 2015. Sampling efficiency is also calculated.

	June				July				August				TOTAL			
Landing Site	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	240	607	18	13.3	292	880	21	13.9	201	591	16	12.6	733	2078	55	13.3
Prince Rupert Yacht Club	201	686	17	11.8	336	1132	17	19.8	194	694	13	14.9	731	2512	47	15.6
Rushbrooke Boat Launch	317	979	18	17.6	430	1426	19	22.6	358	1154	19	18.8	1105	3559	56	19.7
TOTAL	758	2,272	53	14.3	1,058	3,438	57	18.6	753	2,439	48	15.7	2,569	8,149	158	16.3
% of Total Interviews	30%				41%				29%							
Estimated Effort	2,867				3,620				3,272				9,760			
% of Total Effort	29%				37%				34%							
% of Effort Sampled	26%				29%				23%				26%			

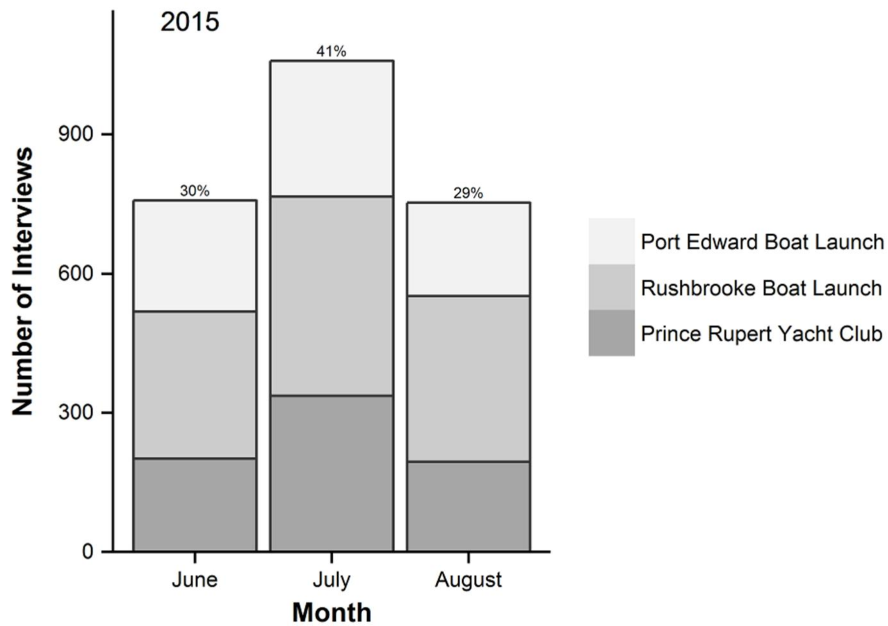


Figure 3. Monthly number of interviews conducted at each of three landing sites, 2015.

Table 4. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2015.

	Trip Type		
Landing Site	Guided	Unguided	% guided
Port Edward Boat Launch	3	730	0.4%
Prince Rupert Yacht Club	413	318	56%
Rushbrooke Boat Launch	458	647	41%
TOTAL	874	1,695	34%

Table 5. Numbers of non-lodge boat trips by month and subarea. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.

	Subarea															
Month	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	TOTAL
June	37	45	9	53	102	85	43	8	221	23	38	28	7	38	23	758
July	53	65	14	56	103	108	114	5	343	38	64	37	2	35	23	1,058
August	30	43	23	43	59	82	60	9	266	23	49	26	6	24	12	753
TOTAL	120	153	45	151	264	276	216	21	830	83	150	91	14	96	58	2,569

Table 6. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea.

	Subarea															
Month	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	TOTAL
June	0	0	0	7	1	3	0	0	0	0	1	0	0	0	0	12
July	0	0	0	16	1	0	0	1	0	0	0	0	0	0	0	18
August	0	0	0	12	3	0	0	1	0	0	0	0	0	0	0	16
TOTAL	0	0	0	35	5	3	0	2	0	0	1	0	0	0	0	46

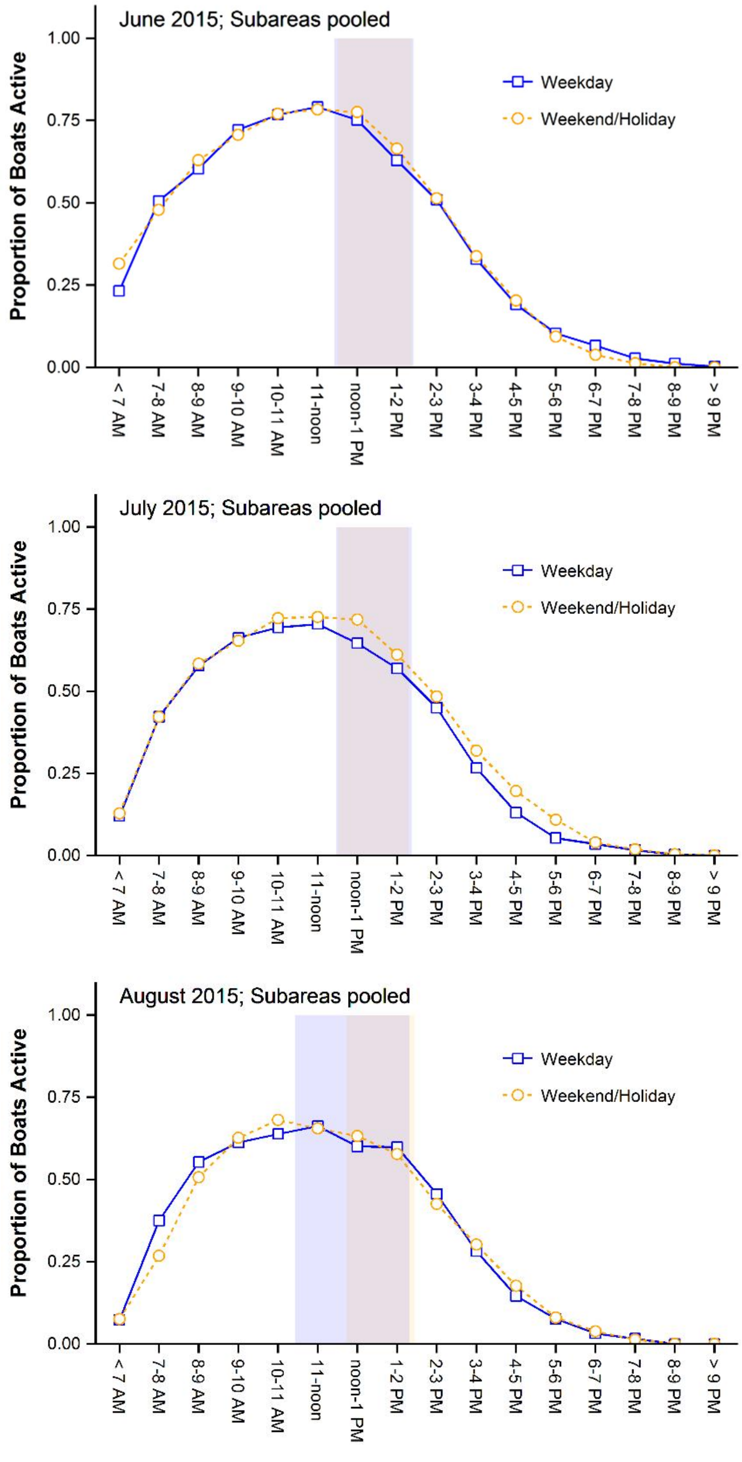


Figure 4. **Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and month, pooled over all subareas. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and month.**

Table 7. Sample sizes for CPE (harvest per effort) estimates by month, for guided and not guided non-lodge boat trips.

		Subarea											
Month	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June	Guided	11	13	1	17	0	2	1	1	61	6	8	8
	Unguided	14	15	6	37	99	76	39	8	113	14	19	7
	Combined	25	28	7	54	99	78	40	9	174	20	27	15
July	Guided	24	22	4	22	3	17	6	1	111	5	30	21
	Unguided	31	19	8	43	107	95	109	4	169	28	21	8
	Combined	55	41	12	65	110	112	115	5	280	33	51	29
August	Guided	21	16	3	14	3	10	4	2	104	3	32	24
	Unguided	5	12	13	33	69	83	60	7	149	16	19	7
	Combined	26	28	16	47	72	93	64	9	253	19	51	31

Table 8. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each month and subarea, for guided and not guided non-lodge boat trips

		Subarea											
Month	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June	Guided	3I	3J	3I,J,K	4A	4A,B,C	4A,B,C	4D,E,F	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
July	Guided	3I	3J	3I,J,K	4A	4A,B,C	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
August	Guided	3I	3J	3I,J,K	4A	4A,B,C	4C	4D,E,F	4D,E,F	4F	4G,H,L	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L

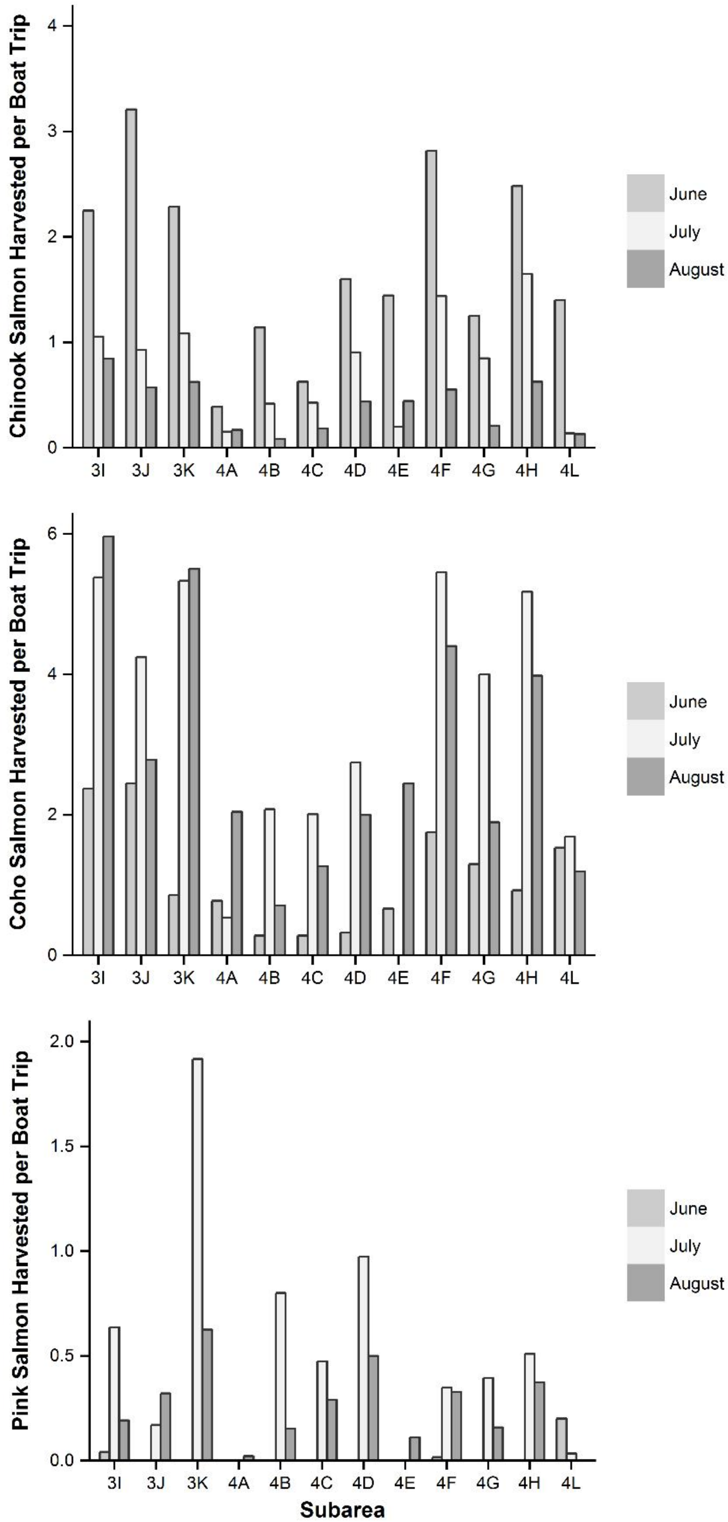


Figure 5. **Average number of Chinook, Coho and Pink salmon harvested per non-lodge boat trip, by month and subarea. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.**

harvested per non-lodge boat trip (HPE) was calculated for each species by month, subarea (Figure 5), and by trip type (Appendix 2).

HPE varied significantly among months (Table 9) for Chinook (χ^2 = 16.3, df = 2, P < 0.001), Coho (χ^2 = 9.4, df = 2, P = 0.009), and Pink salmon (*O. gorbuscha*, χ^2 = 15.5, df = 2, P < 0.001). For Chinook Salmon, median HPE in June was significantly higher than in July or August (Table 9; Figure 5). The opposite was true for Coho and Pink salmon, for which median HPE in June was significantly lower than in July or August (Table 9; Figure 5). There were no significant differences among months for HPE of Chum (*O. keta*) or Sockeye salmon (*O. nerka*), or for any species of groundfish (including Pacific Halibut and Lingcod; Figure 6). For the rockfish species (*Sebastes* spp.), there were no significant differences among months in the number of fish harvested per boat trip. For Canary (*S. pinniger*) and Redstripe rockfish (*S. proriger*), omnibus tests of the effect of month were significant (Canary Rockfish: χ^2 = 6.1, df = 2, P = 0.048; Redstripe Rockfish: χ^2 = 6.3, df = 2, P = 0.042), but none of the *post-hoc* pairwise comparisons were statistically significant after controlling the false discovery rate (i.e., using the Benjamini–Hochberg procedure).

Guided trips had significantly higher HPE than non-guided trips (Table 10) for Chinook (χ^2 = 5.7, df = 1, P = 0.017) and Coho Salmon (χ^2 = 9.8, df = 1, P = 0.002), Pacific Halibut (χ^2 = 17.6, df = 1, P < 0.0001) and Lingcod (χ^2 = 4.5, df = 1, P = 0.033). Guided trips caught 141-172% more salmon and >200% more Pacific Halibut and Lingcod than unguided trips.

Table 9. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among months, by species. Subareas treated as replicates.

	Kruskal-Wallis Statisitics			*Median HPE*		
Species	χ^2	*df*	*P*	*June*	*July*	*August*
Chinook Salmon	16.3	2	0.0003	1.5	0.9	0.4
Coho Salmon	9.4	2	0.009	0.9	3.4	2.2
Pink Salmon	15.5	2	0.0004	0	0.4	0.2
Chum Salmon	2.3	2	0.32	0	0.005	0
Sockeye Salmon	1.6	2	0.46	0	0	0
Pacific Halibut	3.3	2	0.19	1.1	1.0	1.6
Lingcod	0.1	2	0.94	0.2	0.2	0.2
Cabezon	0.5	2	0.77	0	0	0
Dogfish	1.0	2	0.60	0	0	0
Greenling	0.6	2	0.76	0	0	0
Rock Sole	0.7	2	0.69	0	0	0
Black Rockfish	0.9	2	0.65	0	0.01	0.03
Canary Rockfish	6.1	2	0.048	0	0	0
China Rockfish	0.0	2	1.00	0	0	0
Copper Rockfish	2.9	2	0.23	0	0	0
Quillback Rockfish	3.0	2	0.22	0.10	0.09	0.2
Redstripe Rockfish	6.3	2	0.042	0	0	0
Tiger Rockfish	0.7	2	0.71	0	0	0
Yelloweye Rockfish	0.2	2	0.92	0.03	0.04	0.05
Yellowtail Rockfish	0.6	2	0.74	0	0	0

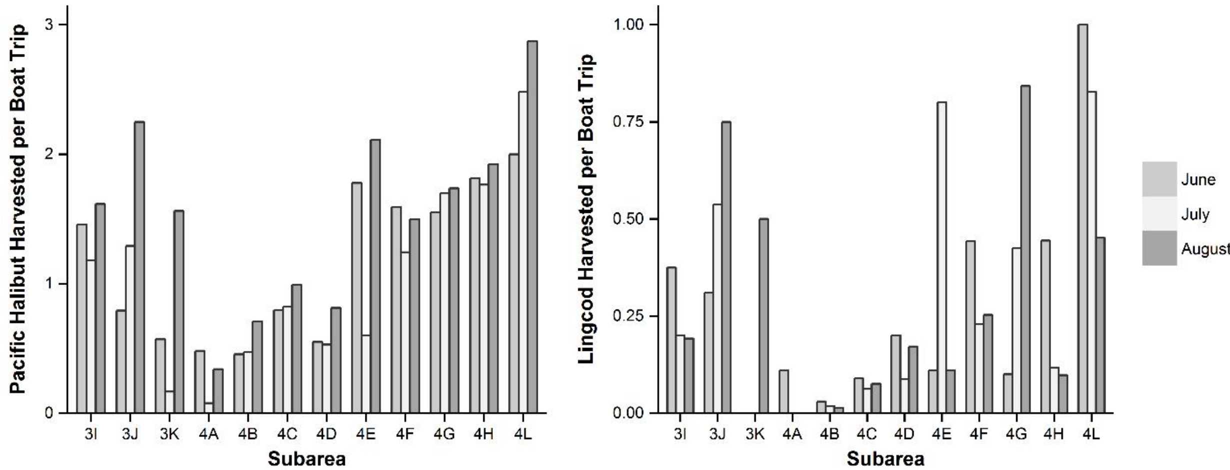


Figure 6. Average number of Pacific Halibut and Lingcod harvested per non-lodge boat trip, by month and subarea. Data are for guided and unguided trips combined. Note that Y axis scale varies between plots.

Table 10. Kruskal-Wallis tests for differences in HPE (fish caught per non-lodge boat trip) between guided and non-guided boats, by species. Months and subareas treated as replicates.

	Kruskal-Wallis Statisitics			*Median HPE*	
Species	χ^2	*df*	*P*	*Guided*	*Not*
Chinook Salmon	5.7	1	0.017	0.9	0.6
Coho Salmon	9.8	1	0.002	3.0	1.8
Pink Salmon	0.0	1	0.84	0.2	0.1
Chum Salmon	0.8	1	0.38	0	0
Sockeye Salmon	1.7	1	0.19	0	0
Pacific Halibut	17.6	1	< 0.0001	1.9	0.8
Lingcod	4.6	1	0.033	0.3	0.1
Cabezon	2.2	1	0.14	0	0
Dogfish	1.0	1	0.33	0	0
Greenling	1.3	1	0.25	0	0
Rock Sole	0.7	1	0.40	0	0
Black Rockfish	0.2	1	0.65	0	0
Canary Rockfish	8.6	1	0.003	0	0
China Rockfish	2.9	1	0.09	0	0
Copper Rockfish	2.3	1	0.13	0	0
Quillback Rockfish	0.7	1	0.42	0.1	0.1
Redstripe Rockfish	3.1	1	0.08	0	0
Tiger Rockfish	3.9	1	0.047	0	0
Yelloweye Rockfish	1.4	1	0.23	0.07	0.03
Yellowtail Rockfish	1.0	1	0.32	0	0

For lodge boat trips, CPE estimates are shown in Table 11. Average numbers of Chinook Salmon retained (1.2 retained per trip) was 131% and 184% higher than the median guided and non-guided non-lodge boat trips, respectively. For Coho Salmon and Lingcod, lodge boats did better than non-guided non-lodge boats (136% and 180%, respectively), but worse than guided non-lodge boats (79% and 80%, respectively). Also, lodge boats caught fewer Pacific Halibut (0.6 fish per boat trip) than either guided (1.9) or non-guided (0.8) non-lodge boats (Tables 10 and 11).

Angler Effort Estimates

Over the three month study period, a total of 24 effort surveys were conducted, including 16 weekday surveys and 8 conducted on weekends/holidays (Table 2). The numbers of lodge boats that were visible during the overflights (Table 12) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts. The remaining boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2015 (Table 12; Figure 7).

Daily non-lodge effort varied significantly between weekdays (median 5.0 trips per day per subarea) and weekends/holidays (11.2 trips per day per subarea; $\chi^2 = 3.8$, df = 1, $P = 0.050$), but there were no significant effect of month (medians: Jun 6.1, July 9.1, August 6.1 trips per day per subarea; $\chi^2 = 0.9$, df = 1, $P = 0.65$).

Total lodge effort in 2015 was 557, 659, and 609 boat trips in June, July and August, respectively. To calculate total (lodge + non-lodge) fishing effort by subarea, we distributed the lodge boat trips among the three subareas in Management Area 3, according to the relative non-

Table 11. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, from census of Area 3 lodges, 2015.

Species	*June*	*July*	*August*	*Total*
Chinook Salmon:				
retained	1.9	1.3	0.3	1.2
released	0.3	0.2	0.06	0.2
Coho Salmon	0.7	2.9	3.4	2.4
Pink Salmon	0.007	0.009	0.002	0.006
Chum Salmon	0.04	0.04	0.02	0.03
Sockeye Salmon	0	0	0.003	0.001
Pacific Halibut	0.5	0.6	0.7	0.6
Lingcod	0.3	0.2	0.3	0.3
Yelloweye Rockfish	0.005	0	0.010	0.005
Other rockfish	0.3	0.4	0.7	0.5
Other groundfish:				
Flounder	0.004	0.002	0	0.002
Kelp Greenling	0.002	0.002	0.010	0.004
Skate	0.002	0	0	0.001
Sole	0.007	0.03	0.002	0.01
True cod	0.09	0.02	0.07	0.06

Table 12. Total numbers of lodge boats that were visible to observers during each of the overflight surveys in 2015.

Flight Date	Lodge Boats Visible to Flight	Flight Date	Lodge Boats Visible to Flight	Flight Date	Lodge Boats Visible to Flight
2 Jun	8	4 Jul	8	2 Aug	12
8 Jun	2	9 Jul	12	5 Aug	11
14 Jun	12	12 Jul	12	11 Aug	11
20 Jun	12	14 Jul	12	16 Aug	11
22 Jun	0	19 Jul	8	17 Aug	3
25 Jun	8	22 Jul	8	21 Aug	0
27 Jun	12	27 Jul	4	27 Aug	8
29 Jun	4	31 Jul	0	31 Aug	6

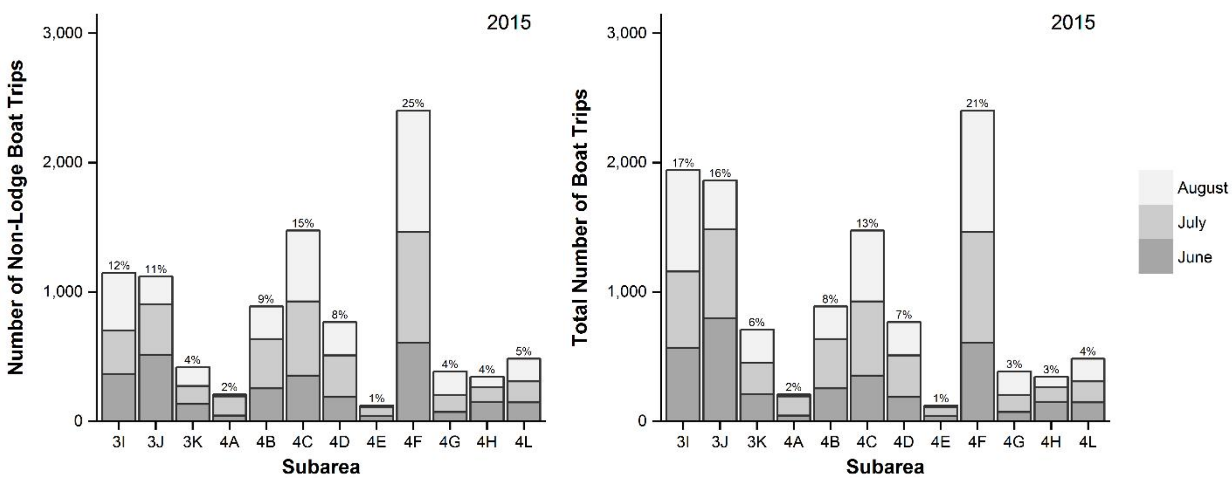


Figure 7. Total estimated numbers of boat trips, by month and subarea. Left panel: estimated monthly non-lodge boat trips. Right panel: Total effort, calculated as the sum of lodge and non-lodge boat trips.

lodge effort in each (Table 13; Figure 7). In all, there were 3,424 (SE = 9) boat trips in June, 4,279 (SE = 10) boat trips in July, and 3,881 (SE = 16) boat trips in August. The full-season Area 3/4 effort estimate for 2015 was 11,585 boat trips (SE = 21).

Catch Estimates

Monthly and full-season estimates were generated of the numbers of fish released and retained, by species (Tables 14-23; Appendix 3). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

Table 13. Lodge, non-lodge and total effort estimates (boat trips per month), by month and subarea.

Month		Subarea 3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
June	Creel Estimate	364.6	513.7	133.9	44.6	254.8	352.4	187.4	40.6	606.8	71.5	150.0	146.9	2,867	9
	Lodge Report	200.7	282.7	73.7										557	
	TOTAL JUNE	565.3	796.3	207.6	44.6	254.8	352.4	187.4	40.6	606.8	71.5	150.0	146.9	3,424	9
	SE	4.3	4.8	1.5	0.7	2.0	2.4	1.7	0.4	4.8	0.7	1.9	2.3		
July	Creel Estimate	335.9	389.4	138.1	148.0	381.4	573.8	322.3	69.6	856.4	132.9	110.8	161.7	3,620	10
	Lodge Report	256.4	297.3	105.4										659	
	TOTAL JULY	592.2	686.7	243.4	148.0	381.4	573.8	322.3	69.6	856.4	132.9	110.8	161.7	4,279	10
	SE	3.3	3.5	1.6	1.4	3.0	3.2	2.6	0.7	6.3	1.2	1.3	2.7		
August	Creel Estimate	447.7	216.2	147.0	15.6	251.0	547.8	257.6	11.1	938.9	181.6	82.5	175.2	3,272	16
	Lodge Report	336.2	162.4	110.4										609	
	TOTAL AUGUST	783.9	378.5	257.4	15.6	251.0	547.8	257.6	11.1	938.9	181.6	82.5	175.2	3,881	16
	SE	6.8	2.9	2.0	0.3	3.3	5.5	4.0	0.2	11.6	2.6	1.2	2.6		
FULL SEASON TOTAL		1,941	1,862	708	208	887	1,474	767	121	2,402	386	343	484	11,585	
FULL SEASON SE		9	7	3	2	5	7	5	1	14	3	3	4	21	

Table 14. Total (lodge + non-lodge) estimated harvest, by month and subarea, for popular sportfish taxa.

Species	Subarea 3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
Chinook Salmon	2,343	3,149	843	43	471	567	704	77	3,461	240	607	251	12,756	411
Coho Salmon	7,681	4,798	2,430	146	1,044	1,947	1,462	54	9,868	969	1,041	708	32,148	709
Pink Salmon	322	137	360	0	343	431	443	1	618	81	87	35	2,858	165
Chum Salmon	135	0	34	0	3	12	11	0	39	4	11	0	249	37
Sockeye Salmon	16	0	12	0	0	22	3	0	28	10	4	0	95	22
Pacific Halibut	2,196	1,883	429	38	474	1,293	483	137	3,437	652	626	1,198	12,848	321
Lingcod	450	820	112	5	18	109	110	61	702	216	88	360	3,051	190
other groundfish *	122	0	131	1	18	46	43	0	84	0	20	28	492	44
Rockfish †	670	1,335	171	20	220	393	586	18	1,807	459	104	929	6,712	269

* includes Cabezon, Dogfish, Starry Flounder, Greenling, Ratfish, Rock Sole, and unknown groundfish

† includes Black, Canary, China, Copper, Quillback, Redstripe, Tiger, Yelloweye, Yellowtail and unknown rockfish

Table 15. Total (lodge + non-lodge) estimated Chinook Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*TOTAL*	*SE*
June														
Retained	1126.0	2260.7	420.0	17.3	290.9	221.4	299.9	58.6	1708.7	89.4	372.2	205.7	7,071	372
Released	19.1	622.3	48.0	4.1	61.8	189.8	103.1	0	439.4	3.6	33.3	9.8	1,534	222
July														
Retained	709.7	723.3	299.7	22.8	159.5	245.9	291.5	13.9	1232.6	112.8	182.5	22.3	4,016	135
Released	112.6	122.6	42.4	2.3	111.0	5.1	42.0	0	107.0	16.1	32.6	0	594	70
August														
Retained	506.9	165.3	123.0	2.7	20.9	100.1	112.7	4.9	519.6	38.2	51.8	22.6	1,669	110
Released	0	49.4	0	0	0	5.9	0	1.2	33.4	9.6	3.2	0	103	19
TOTAL														
Retained	2,343	3,149	843	43	471	567	704	77	3,461	240	607	251	12,756	411
Released	132	794	90	6	173	201	145	1	580	29	69	10	2,231	234
Catch	2,474	3,944	933	49	644	768	849	79	4,041	270	676	260	14,987	473

Table 16. Total (lodge + non-lodge) estimated Coho Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*TOTAL*	*SE*
June														
Retained	1014.2	1472.7	134.4	34.7	72.1	99.4	60.9	27.1	1063.6	93.0	138.9	225.3	4,436	383
Released	76.0	35.4	0	0	33.5	4.5	42.2	9.0	94.2	7.2	0	0	302	65
July														
Retained	2644.4	2418.0	1077.3	79.7	794.0	1152.7	885.6	0	4670.4	531.7	573.6	273.2	15,101	389
Released	836.6	417.9	46.0	18.2	260.0	71.7	238.2	417.8	394.6	44.3	67.4	0	2,813	480
August														
Retained	4022.4	907.6	1218.5	31.9	177.8	695.1	515.2	27.1	4134.3	344.1	328.5	209.2	12,612	453
Released	34.4	123.5	0	0.7	0	35.3	20.1	0	181.8	19.1	1.6	5.7	422	74
TOTAL														
Retained	7,681	4,798	2,430	146	1,044	1,947	1,462	54	9,868	969	1,041	708	32,148	709
Released	947	577	46	19	294	112	301	427	671	71	69	6	3,537	490
Catch	8,628	5,375	2,476	165	1,337	2,059	1,762	481	10,539	1,039	1,110	713	35,685	862

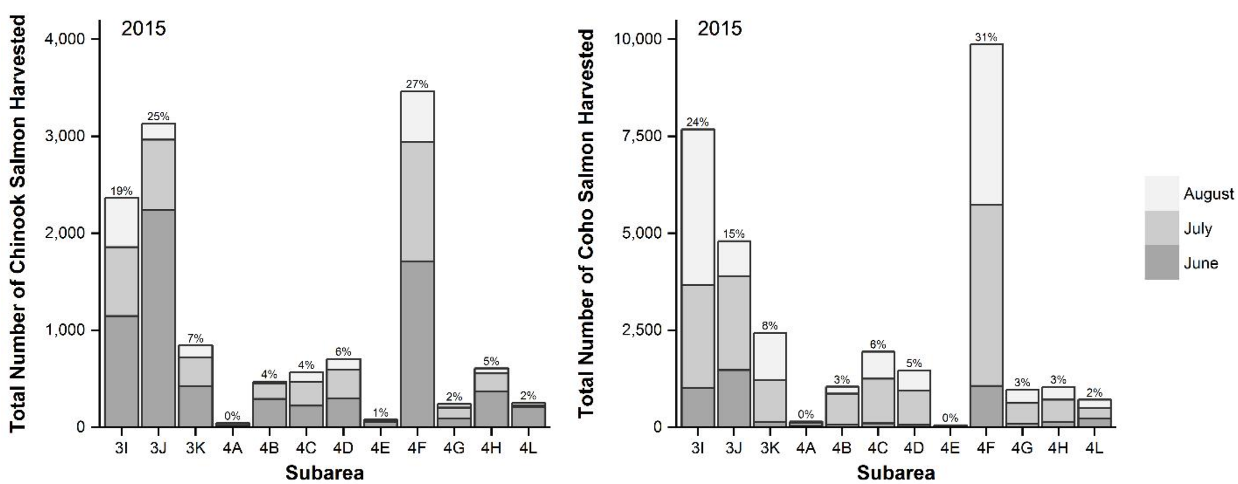


Figure 8. Total (lodge + non-lodge) estimated Chinook and Coho salmon harvest, by month and subarea. Note that Y axis scale varies between plots.

An estimated 12,756 (SE = 411) Chinook Salmon were retained by anglers in Areas 3 or 4 during 2015 the study period (Table 15; Figure 8). The majority of harvest came from subareas 4F (27%), followed by 3J (25%) and 3I (18%). Harvest declined over the study period: it was significantly higher in June (55% of harvest) and July (31% of harvest) than in August (13% of harvest; $\chi^2 = 12.0$, df = 2, $P = 0.002$). During the study, the number of released Chinook Salmon was estimated at 2,231 (SE = 234), or 15% of the total catch.

The species that was harvested in the greatest numbers during the 2015 study period was, by far, Coho Salmon (32,148 fish retained, SE = 709; Table 16; Figure 8). The majority of harvest came from subareas 4F (31%) and 3I (24%), followed by 3J (15%) and 3K (8%). In June, 14% of the total Coho Salmon harvest was recorded, significantly less than in July or August (47% and 39% of the total harvest, respectively; $\chi^2 = 10.0$, df = 2, $P = 0.007$). During the study period, an estimated 3,537 (SE = 490) Coho Salmon were released by anglers (10% of the total catch).

Pink salmon was the least likely salmonid to be retained in 2015 (Table 17). We estimate that 9,607 (SE = 725) fish were caught, of which 30% were retained (2,858 fish, SE = 165), and 70% were released (6,749 fish, SE = 706). Harvest of Pink Salmon was significantly higher in July (65% of total) and August (33% of total) than it was in June (2% of total; $\chi^2 = 28.6$, df = 2, $P < 0.0001$). Very few Chum (249 fish, SE = 37) or Sockeye (95 fish, SE = 22) salmon were harvested in Area 3 and 4 during the study period (Tables 18 and 19).

An estimated 12,848 (SE = 321) Pacific Halibut were harvested in Areas 3 or 4 during the 2015 study period (Table 20; Figure 9). Most of the harvest occurred in subarea 4F (27%), followed by 3I (17%) and 3J (15%). Harvest increased over the study period, from 28% in June, 32% in July, and 40% in August, though the differences were not statistically significant ($\chi^2 = 2.2$, df = 2, $P = 0.33$). We estimated that 4,475 (SE = 324) Pacific Halibut were released (26% of the total catch).

From June to August 2015, an estimated 3,051 (SE = 190) Lingcod were harvested, with no discernable temporal pattern (Table 21; Figure 9). The harvest occurred largely in subarea 3J (27%), 4F (23%), 3I (15%) and 4L (12%). Only 3% of the total number of captured Lingcod were released.

During the study period, an estimated total of 492 other groundfish (SE = 44) were retained (Table 22), including Cabezon (*Scorpaenichthys marmoratus*), and various species of dogfish, skates, cod, flounders, soles, and others (120 fish, SE= 32, were released). Rockfish harvest was estimated at 6,712 fish (SE = 269; Table 23), including Black (*Sebastes melanops*), Canary, China (*S. nebulosus*), Copper (*S. caurinus*), Quillback (*S. maliger*), Redstripe, Tiger (*S. nigrocinctus*), Yelloweye (*S. ruberrimus*), Yellowtail (*S. flavidus*) and other rockfish species (416 fish, SE = 63, were released).

Table 17. Total (lodge + non-lodge) estimated Pink Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
June														
Retained	19.2	0	0	0	0	0	0	0	10.5	0	0	29.4	59	18
Released	0	0	0	0	0	0	0	0	3.5	0	0	0	3	3
July														
Retained	216.1	67.2	267.5	0	305.1	271.5	313.9	0	299.7	52.4	56.5	5.6	1,856	134
Released	164.9	721.9	759.4	0	374.5	517.5	266.2	0	978.7	177.2	60.8	0	4,021	634
August														
Retained	86.4	69.8	92.2	0.3	38.3	159.0	128.8	1.2	308.0	28.7	30.7	0	944	94
Released	292.7	239.3	45.9	3.3	125.5	541.9	309.9	34.5	842.4	181.6	106.8	0	2,724	311
TOTAL														
Retained	322	137	360	0	343	431	443	1	618	81	87	35	2,858	165
Released	458	961	805	3	500	1,059	576	35	1,825	359	168	0	6,749	706
Catch	779	1,098	1,165	4	843	1,490	1,019	36	2,443	440	255	35	9,607	725

Table 18. Total (lodge + non-lodge) estimated Chum Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
June														
Retained	53.4	0	0	0	0	0	0	0	7.0	3.6	0	0	64	18
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July														
Retained	81.3	0	15.3	0	3.5	0	11.2	0	24.5	0	10.9	0	147	30
Released	12.2	0	0	0	0	0	5.6	0	6.1	0	0	0	24	8
August														
Retained	0	0	19.2	0.3	0	11.8	0	0	7.4	0	0	0	39	11
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL														
Retained	135	0	34	0	3	12	11	0	39	4	11	0	249	37
Released	12	0	0	0	0	0	6	0	6	0	0	0	24	8
Catch	147	0	34	0	3	12	17	0	45	4	11	0	273	38

Table 19. Total (lodge + non-lodge) estimated Sockeye Salmon catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
June														
Retained	16.3	0	0	0	0	0	0	0	7.0	0	0	0	23	13
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July														
Retained	0	0	12.4	0	0	10.2	2.8	0	6.1	0	2.2	0	34	12
Released	0	0	0	0	41.6	0	0	0	0	0	0	0	42	31
August														
Retained	0	0	0	0	0	11.8	0	0	14.8	9.6	1.6	0	38	12
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL														
Retained	16	0	12	0	0	22	3	0	28	10	4	0	95	22
Released	0	0	0	0	42	0	0	0	0	0	0	0	42	31
Catch	16	0	12	0	42	22	3	0	28	10	4	0	136	38

Table 20. Total (lodge + non-lodge) estimated Pacific Halibut catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*TOTAL*	*SE*
June														
Retained	688.8	527.7	99.1	21.5	115.8	280.1	103.1	72.2	966.0	110.8	272.2	293.9	3,551	204
Released	212.7	106.3	0	11.6	23.2	117.5	4.7	36.1	477.8	60.8	88.9	431.0	1,570	224
July														
Retained	573.2	727.0	33.2	11.4	180.3	471.3	171.0	41.8	1064.4	225.5	195.5	401.4	4,096	151
Released	262.6	256.5	0	0	48.5	128.1	30.8	83.6	406.8	80.6	63.0	256.5	1,617	190
August														
Retained	934.2	628.3	296.7	5.3	177.8	541.9	209.3	23.4	1406.5	315.5	158.6	503.1	5,201	197
Released	86.1	262.5	18.4	1.0	17.4	64.8	68.4	3.7	348.9	172.1	63.1	180.9	1,287	137
TOTAL														
Retained	2,196	1,883	429	38	474	1,293	483	137	3,437	652	626	1,198	12,848	321
Released	561	625	18	13	89	310	104	123	1,233	313	215	868	4,475	324
Catch	2,758	2,508	447	51	563	1,604	587	261	4,670	965	841	2,067	17,322	456

Table 21. Total (lodge + non-lodge) estimated Lingcod catch (retained and released), by month and subarea, shown separately for released and retained fish.

	Subarea													
Month / Catch Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*TOTAL*	*SE*
June														
Retained	215.2	250.9	0	5.0	7.7	31.6	37.5	4.5	268.5	7.2	66.7	146.9	1,042	132
Released	15.2	0	0	0.8	0	0	0	0	7.0	0	0	0	23	13
July														
Retained	103.4	321.7	0	0	6.9	35.9	28.0	55.7	195.7	56.4	13.0	133.8	951	93
Released	0	0	0	2.3	6.9	0	0	0	0	0	0	0	9	4
August														
Retained	131.3	247.3	112.1	0	3.5	41.2	44.3	1.2	237.5	153.0	8.1	79.1	1,059	100
Released	0	15.4	0	0	0	11.8	16.1	0	3.7	0	0	0	47	16
TOTAL														
Retained	450	820	112	5	18	109	110	61	702	216	88	360	3,051	190
Released	15	15	0	3	7	12	16	0	11	0	0	0	79	21
Catch	465	835	112	8	25	121	126	61	712	216	88	360	3,130	192

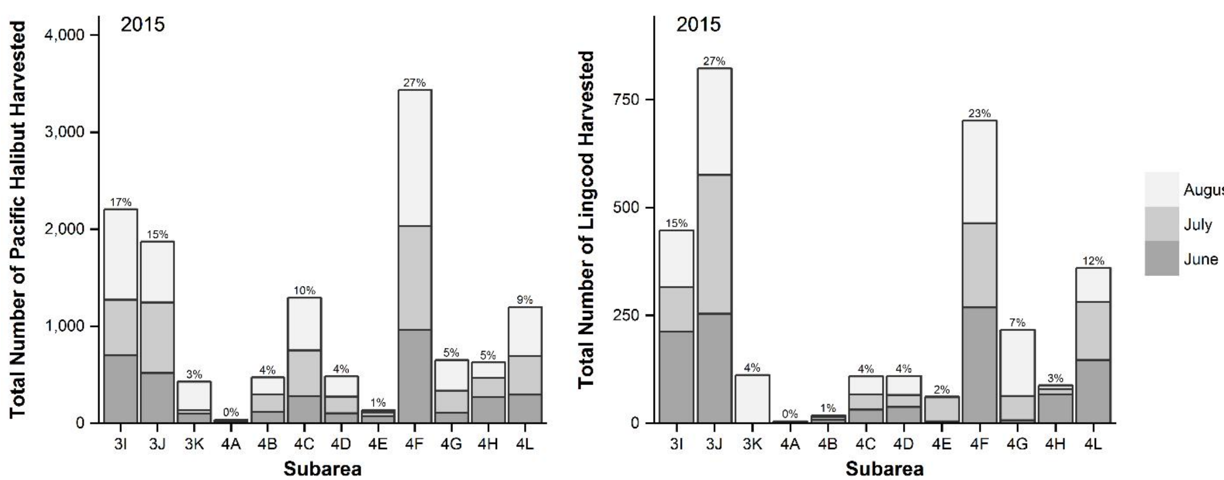


Figure 9. Total (lodge + non-lodge) estimated Pacific Halibut and Lingcod harvest, by month and subarea. Note that Y axis scale varies between plots.

Table 22. Total (lodge + non-lodge) estimated catch (retained and released) of other groundfish, by month and subarea, shown separately for released and retained fish. Included are Cabezon, Dogfish, Starry Flounder, Greenling, Ratfish, Rock Sole, and other miscellaneous groundfishes.

	Subarea													
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
June														
Retained	73.2	0	0	0	7.7	18.1	4.7	0	24.4	0	11.1	0	139	20
Released	30.4	0	0	0	0	4.5	14.1	0	0	0	0	0	49	26
July														
Retained	48.8	0	36.8	0	10.4	10.2	22.4	0	18.4	0	2.2	0	149	26
Released	0	0	0	0	0	20.5	0	0	3.1	0	0	0	24	12
August														
Retained	0	0	93.9	0.7	0	17.7	16.1	0	40.8	0	6.5	28.3	204	30
Released	0	0	0	0	7.0	11.8	8.1	0	3.7	9.6	1.6	5.7	47	13
TOTAL														
Retained	122	0	131	1	18	46	43	0	84	0	20	28	492	44
Released	30	0	0	0	7	37	22	0	7	10	2	6	120	32
Catch	152	0	131	1	25	83	65	0	90	10	21	34	612	55

Table 23. Total (lodge + non-lodge) estimated rockfish catch (retained and released), by month and subarea, shown separately for released and retained fish. Included are Black, Canary, China, Copper, Quillback, Redstripe, Tiger, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea													
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL	SE
June														
Retained	134.5	282.2	33.9	18.2	66.9	99.4	107.8	4.5	610.3	32.2	27.8	215.5	1,633	130
Released	15.2	0	0	0	2.6	31.6	28.1	0	38.4	0	0	0	116	29
July														
Retained	250.3	389.4	0	0	62.4	199.8	213.0	0	532.2	120.8	36.9	312.2	2,117	158
Released	30.5	0	0	0	24.3	25.6	16.8	0	21.4	12.1	0	5.6	136	32
August														
Retained	284.7	663.9	136.7	1.7	90.6	94.2	265.7	13.6	664.3	305.9	38.8	401.4	2,961	175
Released	0	0	0	0	0	5.9	20.1	0	40.8	76.5	3.2	17.0	164	46
TOTAL														
Retained	670	1,335	171	20	220	393	586	18	1,807	459	104	929	6,712	269
Released	46	0	0	0	27	63	65	0	101	89	3	23	416	63
Catch	715	1,335	171	20	247	457	651	18	1,907	547	107	952	7,127	276

Biosampling

During interviews, observers were told about 3,496 retained Chinook Salmon, but were only able to biosample 996 fish. The observed sex ratio was 54:46 (male:female), and most fish were red-fleshed (97%; Table 24). The median fork length of biosampled Chinook Salmon (n = 903) was 74 cm (95% of observations were between 53.7 and 98 cm; Figure 10). Of the biosampled Chinook Salmon, 13% were adipose-clipped and 12% had their heads collected and labelled for submission to the Salmon Head Recovery Program. Scale samples were obtained from 92% of the biosampled Chinook Salmon (Table 24).

For Coho Salmon, observers were told about 9,162 retained fish, but biosampled only 23 fish. Nine fish were sexed, and the observed sex ratio was 2:1 (male:female; Table 24). The median

Table 24. Biosampling results.

	Chinook Salmon		Coho Salmon		Pacific Halibut	
	Number	Percent	Number	Percent	Number	Percent
Number Examined	996		23		734	
Subarea						
3I	47	5%	1	4%	35	5%
3J	68	7%	1	4%	44	6%
3K	16	2%			4	1%
4A	13	1%			6	1%
4B	103	11%	1	4%	58	8%
4C	47	5%			75	10%
4D	103	11%	1	4%	39	5%
4E	10	1%			14	2%
4F	406	44%	11	48%	292	41%
4G	35	4%	3	13%	48	7%
4H	72	8%	5	22%	52	7%
4L	9	1%			51	7%
Sex						
Male	232	54%	6	67%	1	33%
Female	196	46%	3	33%	2	67%
Undeterminable	568		14		731	
Unknown						
Trip Type						
Guided	457	46%	8	35%	362	49%
Unguided	539	54%	15	65%	372	51%
Adipose Clips						
Clipped	133	13%	4	17%		
Unmarked	862	87%	19	83%		
Unknown	1					
Head Tagged						
Yes	119	12%	4	17%		
No	877	88%	19	83%		
Scale Samples						
Yes	913	92%				
No	83	8%				
Flesh Colour						
Red	971	97%				
White	24	2%				
Marble	1	0%				
Unknown						

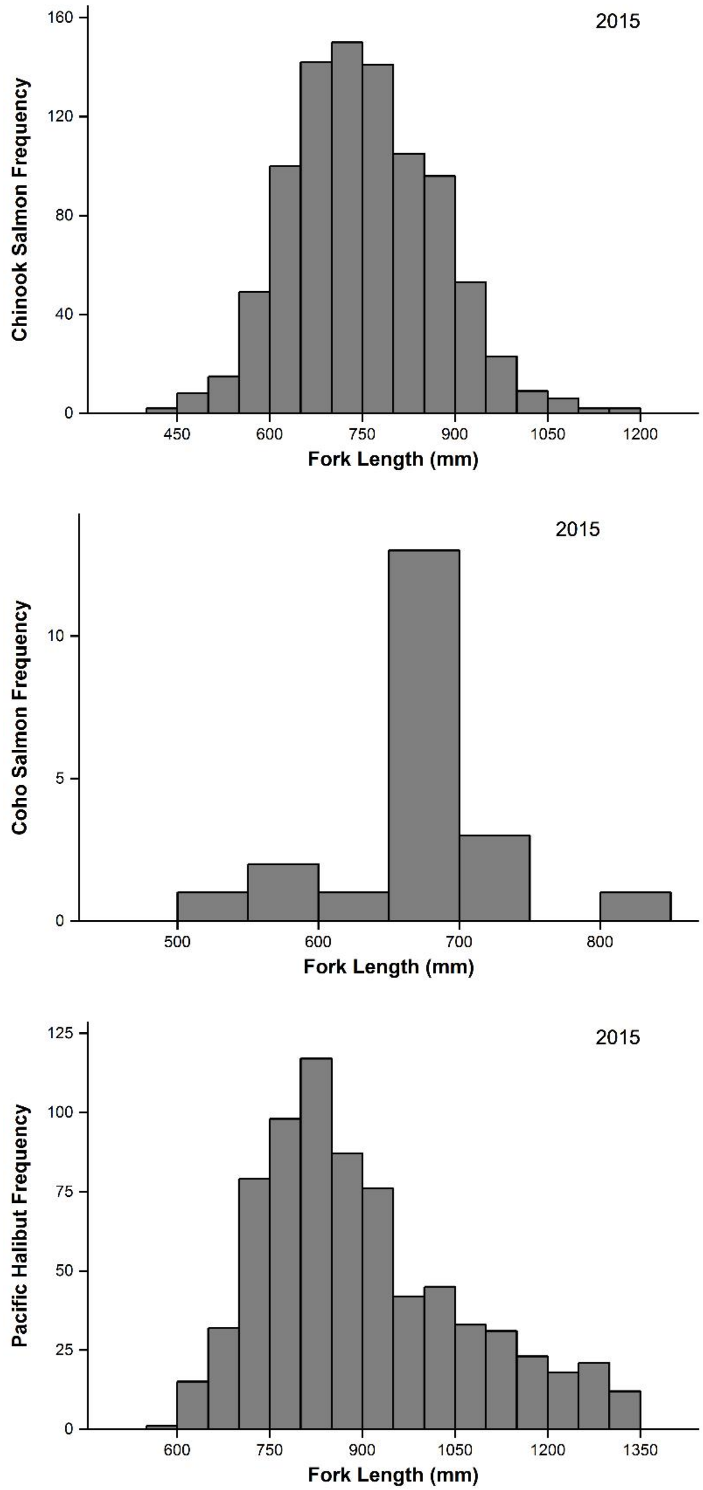


Figure 10. Length-frequency distributions of biosampled Chinook (n = 903) and Coho (n = 21) salmon, and Pacific Halibut (n = 730), 2015. Note that Y axis scale varies among plots.

Coho Salmon fork length (n = 21) was 66.5 cm (95% of observations were between 54 and 80 cm; Figure 10). For Coho Salmon, 23 were biosampled, of which 4 had clipped adipose fins (all four heads were collected and labelled for submission to the Salmon Head Recovery Program).

Of the 3,798 Pacific Halibut that were interviewed anglers reported as retained, 734 were observed (Table 24). Three fish were sexed (1:2 male:female), and 730 were measured. The median size was 86 cm (95% of observations were between 65 and 128 cm; Figure 10). Average net and round weights, calculated from average lengths using regression formulas, were 6.5 kg (14.4 lb) and 8.7 kg (19.1 lb) in Area 3, and 7.6 kg (16.8 lb) and 10.1 kg (22.2 lb) in Area 4, respectively (Table 25).

Harvested Halibut Biomass

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 92.8 tons (29.5 t in Area 3 and 63.4 t in Area 4); 123.0 tons round (39.0 t in Area 3 and 83.9 t in Area 4; Table 25) [1].

Table 25. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2015.

	Stat Area 3				*Stat Area 4*			
	Net Weight		*Round Weight*		*Net Weight*		*Round Weight*	
	kg	*lb*	*kg*	*lb*	*kg*	*lb*	*kg*	*lb*
Creel Interview (Non-Lodge) Average Weights								
Guided	7.0	15.3	9.2	20.3	8.5	18.7	11.2	24.8
Unguided	5.8	12.8	7.7	16.9	6.8	15.0	9.0	19.9
Overall	6.5	14.4	8.7	19.1	7.6	16.8	10.1	22.2
Estimated Non-lodge Harvest (number of fish)								
Overall	3,378				8340			
Estimated Lodge Harvest (number of fish)								
Overall	1,130				0			
Biomass Harvested								
Non-lodge	22,082	48,683	29,245	64,475	63,377	139,722	83,935	185,045
Lodge	7,386	16,284	9,782	21,566	0	0	0	0
Overall	29,469	64,967	39,028	86,041	63,377	139,722	83,935	185,045

[1] There are 2204.62 pounds in a metric ton (t), thus, the total harvested Pacific Halibut net biomass was 204,689 lbs, (64,967 lbs in Area 3 and 139,722 lbs in Area 4); 271086 lbs round (86,041 lbs in Area 3 and 185,045 lbs in Area 4).

DISCUSSION

Creel Estimates

Recreational fishing effort, salmon CPUE, and salmon catch in Areas 3 & 4 have been increasing over the last 20 years (Table 26). Using creel data collected since 1995, we found seven years for which effort data were collected in June, July and August (see Van Tongeren and Winther 2010), and limited analyses to those three months (1995, 1996, 2001, 2002, 2008, 2009, 2015). Meta-analysis revealed (Figure 11) that fishing effort has increased steadily (by 236%) from 4,911 trips in 1995 to 11,585 boat trips in 2015 (an average increase of 318 trips per year). Over the same period, Chinook Salmon harvest has increased by 770%, from 1,657 to 12,756 fish (an average increase of 500 Chinook Salmon per year; Table 27, Figure 11). Similarly, Coho Salmon has increased by 1216%, from 2,644 fish in 1995 to over 30,000 fish (an average increase of 1728 Coho Salmon per year), though harvest may have levelled-off in recent years (in the data available, peak June-August harvest was in 2009 at 37,188 fish, Table 27, Figure 11), but it is difficult to draw firm conclusions without access to the 2010-2014 results. Given that harvest has increased more rapidly that effort, it follows that CPE must have also been increasing

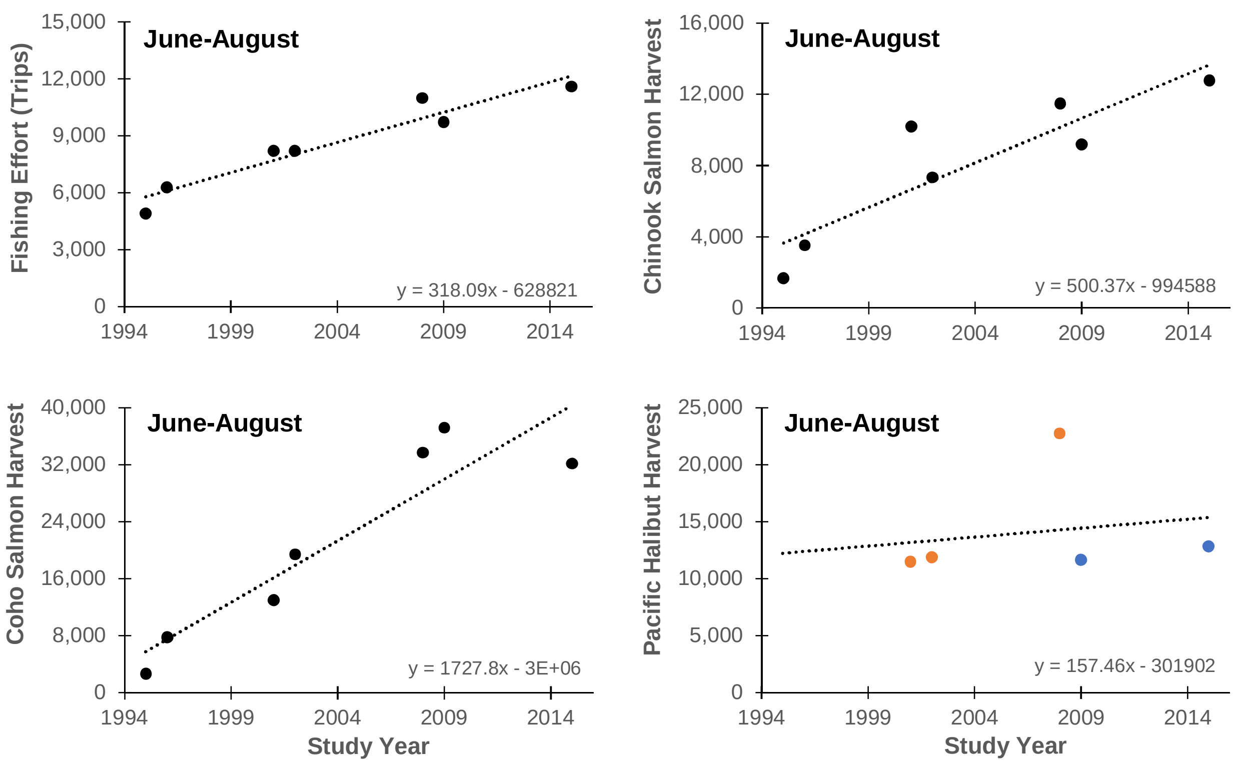


Figure 11. Fishing effort, and harvest of Chinook Salmon, Coho salmon and Pacific Halibut from 1995 to 2015. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Linear regression equations are shown in the lower right of each panel. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).

Table 26. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2015, as estimated using creel survey methods.

	Year																		
Month	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
May	965	391	--	--	--	--	449	960	--	--	--	983	--	--	--	--	--	--	--
June	1,590	2,490	--	--	--	--	3,801	3,463	--	--	--	4,696	3,043	--	--	--	--	--	3,424
July	1,394	2,385	--	1,307	--	1,842	2,588	2,892	--	--	--	3,750	3,804	--	--	--	--	--	4,279
Aug	1,927	1,400	--	1,351	--	1,327	1,808	1,843	--	2,593	--	2,529	2,865	--	--	--	--	--	3,881
Sept	864	488	--	504	--	954	540	676	--	758	--	394	817	--	--	--	--	--	--
TOTAL	6,740	7,154	--	3,162	--	4,123	9,186	9,834	--	3,351	--	12,352	10,529	--	--	--	--	--	11,585

Table 27. Monthly Chinook Salmon, Coho Salmon and Pacific Halibut harvest in the Area 3 and 4 recreational fishery from 1995 to 2015, as estimated using creel survey methods.

	Year																		
Month	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Chinook Salmon Harvest																			
May	141	45	--	--	--	--	509	522	--	--	--	474	--	--	--	--	--	--	--
June	907	2,165	--	--	--	--	6,630	4,028	--	--	--	7,708	4,797	--	--	--	--	--	7,071
July	505	1,097	--	254	--	1,611	3,260	2,822	--	--	--	3,080	3,873	--	--	--	--	--	4,016
Aug	245	257	--	35	--	108	299	474	--	1,207	--	688	503	--	--	--	--	--	1,669
Sept	15	62	--	6	--	93	61	59	--	203	--	20	4	--	--	--	--	--	--
TOTAL	1,813	3,626	--	295	--	1,812	10,759	7,905	--	1,410	--	11,970	9,177	--	--	--	--	--	12,756
Coho Salmon Harvest																			
May	0	0	--	--	--	--	0	0	--	--	--	0	--	--	--	--	--	--	--
June	335	474	--	--	--	--	1,771	1,332	--	--	--	3,913	3,686	--	--	--	--	--	4,436
July	402	3,565	--	1,937	--	568	4,736	9,798	--	--	--	15,982	17,043	--	--	--	--	--	15,101
Aug	1,907	3,771	--	1,500	--	2,125	6,488	8,292	--	15,647	--	13,791	16,459	--	--	--	--	--	12,612
Sept	1,115	1,417	--	79	--	790	1,006	1,010	--	3,215	--	733	3,136	--	--	--	--	--	--
TOTAL	3,759	9,227	--	3,516	--	3,483	14,001	20,432	--	18,862	--	34,419	40,324	--	--	--	--	--	32,148
Pacific Halibut Harvest																			
May	--	--	--	--	--	--	281	467	--	--	--	1,435	--	--	--	--	--	--	--
June	--	--	--	--	--	--	4,872	3,912	--	--	--	8,367	3,029	--	--	--	--	--	3,551
July	--	--	--	--	--	3,963	3,535	4,716	--	--	--	8,507	4,466	--	--	--	--	--	4,096
Aug	--	--	--	1,801	--	2,530	3,085	3,247	--	8,178	--	5,865	4,142	--	--	--	--	--	5,201
Sept	--	--	--	328	--	1,224	222	615	--	1,812	--	774	1,205	--	--	--	--	--	--
TOTAL	0	0	--	2,129	--	7,717	11,995	12,957	--	9,990	--	24,948	12,842	--	--	--	--	--	12,848

over the past 20 years. CPE is measured in 'catch per boat trip', thus CPE can increase if there are more anglers per boat, if more hours are spent fishing per boat trips, if more efficient equipment is used, or if fish abundances increase such that they are caught more readily. It is outside the scope of this report to determine whether the current salmon harvest levels are appropriate for our present-day stock abundances.

Halibut harvest did not show a strong among-year trend (Figure 11). While fishing effort increased, the total harvest of Pacific Halibut in recent years (2009, 2015) is very similar to that from 2001-2002; Table 27). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 11).

Additional data may be available to add to this meta-analysis, but DFO's creel results from 2010-2014 have not been published. Given recent funding cuts and staff reductions at DFO, it is not surprising that publication of these data has not been prioritized. Nevertheless, the results from the creel surveys should be made available, especially since they might elucidate how effort and harvest have changed in the recent years. Perhaps it may be within the purview of external funding agencies to support the preparation of a manuscript report that would make these data public.

Biosampling

The fish that were biosampled in 2015 were not a random selection of the observed catch. This was evident from the mismatch in the fin-clip incidence rates between the interview and biosample datasets. During interviews, observers saw 2,318 Chinook Salmon, of which 144 (6.2%) were indicated as adipose-clipped; and 6,427 Coho Salmon, of which 13 (0.2%) were indicated as clipped. Apparently, the observers preferentially biosampled fish when clipped fish were available, because 13.4% and 17.4% of the biosampled Chinook and Coho salmon were adipose-clipped, respectively. This bias makes sense, since interviewers were told that collecting heads from clipped fish was a priority task, and while collecting heads, it made sense to do a complete biosample. It is possible that the clip-incidence rates in the interview data may be under-reported (as the observed fish were not always handled, or viewed in close enough proximity to be inspected for marks), so the exact magnitude of the biosampling bias is not known. Nevertheless, this bias will most likely have an impact on the reliability of any stock-composition analyses that are based on the biosampled data.

It may be possible to reduce the biosampling bias by changing the layout of the datasheet. The interview datasheet, adapted from the Strait of Georgia creel survey, prompts the observer to ask about the clip status of fish before asking for permission to take biosamples from catch. If the datasheet, and therefore the sequence of interview questions, is re-ordered, it may be possible to prevent the preferential sampling of clipped salmon.

Precision of the Results

In 2015, we met our precision goal, to estimate total harvests within 25% of the true value 19 times out of 20, for all the main taxa. For example, Chinook Salmon, Coho Salmon and Pacific Halibut harvest estimates were precise to within 4-6%. Rockfish estimates were precise to within 8%; Pink Salmon and Lingcod estimates were precise to within 11-12%; and harvest estimate for 'other groundfish' was precise to within 18%. By contrast, there were two taxa for which precision goals were not met: the Chum Salmon harvest estimates were precise to within 29%, and those for Sockeye Salmon were precise to within 45%.

The precision of our harvest estimates results from the variance (or standard error), which, in turn, results from two factors: 1) variability in CPE; and 2) sampling error. When sample sizes are large, variability in CPE is the main factor affecting precision. Catch rates tend to follow a negative binomial distribution, where most catches are of zero fish; and the larger the catch, the rarer the event. If the fish were uniformly distributed and anglers had equal experience and ability, there would be considerably less variability in the CPE estimates. However, day-to-day changes in abundance of the target species, fishing effort, weather or fishing conditions typically result in a wide range of outcomes for any given fishing trip. This degree of variability translates into the size of the confidence limits around the catch estimates for each species.

Sampling error is the other main source of estimation error. As with any scientific study that involves sampling, the confidence you have in your final estimate is greater when a larger proportion of population has been sampled. With catches expected to be widely variable, it follows that the precision of estimates drawn from a small number of samples would be low. One solution is to pool data among categories, but this is not ideal since we know a priori that catch rates differ among, for example, months. The other solution is to increase interviewing and survey effort. However, personnel and budget limitations restrict most recreational creel surveys to sample less than 20% of the total fishing effort. In this study, we sampled 26% of the non-lodge fishing effort, so there is not much reasonable room for increase survey effort.

Given the relatively good precision for most of the taxa surveyed, it appears that our level of survey effort was adequate to meet our goal in most cases. It was undoubtedly the variability in CPE for Chum and Sockeye salmon, coupled with relatively small total harvests, that resulted in the missed precision goals for these two species – both factors over which we have no control.

Accuracy of the Results

The accuracy the results depends on the quality of data provided by the anglers to the interviewers. Each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 2,318 of the 3,496 Chinook Salmon (66%) and 6,427 of the 9,162 Coho Salmon (70%) reported as kept during angler interviews were recorded as observed by the interviewer. While not every fish can be observed[2], and these observation rates are indeed quite high, each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals.

The adipose clip-incidence rates derived from the interview data are also potentially inaccurate. When anglers allow interviewers to observe their fish, they do not necessarily allow their catch

[2] Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, or boat launch traffic congestion causing tension, among other reasons.

to be handled, or even seen from a close distance. So, while quantity and species identifications can be verified this way, it is not always possible to examine the fish individually for the presence or absence of adipose fins. Thus the incidence of clips in the interview data sometimes relies upon the anglers' participation.

In this study, the overflights are another potential source of inaccuracy. In general, we have no estimate of the proportion of angling boats that are missed during overflights. Any missed boats would contribute to the underestimation of recreational fishing effort. We do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands) where overflights cannot see. We also do not know how visibility affects observer efficiency. We recommend that a simple study be conducted in order to provide an estimate of the proportion of angling vessels that are recorded along the standard flightpath. The study could involve two simultaneous aerial surveys, one following the standard flightpath and another doing a more in-depth survey that covers more ground.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. The method follows these steps: 1) boats are counted during overflights, without knowledge of whether or not they are associated with a lodge; 2) the lodge reports how many boats were fishing on the date of the overflight; 3) DFO communicates with each of the lodges to determine the approximate proportion of lodge vessels that were available to be counted during the overflight; and 4) the available lodge boats are subtracted from the overflight counts in order to obtain the number of non-lodge boats that were fishing. There are several problems with this method. One problem is that we do not know whether all of the lodge boats were observed. For example, some may have been outside of the survey area (e.g., up Portland Inlet). Any if any were indeed missed, then non-lodge fishing effort would be underestimated. Another problem is that we currently have no estimate of the accuracy of the lodge's reported boat activity or the calculated proportion of lodge vessels that were available during the survey. Errors in these values would have direct impact on the accuracy of the estimates of non-lodge fishing effort.

RECOMMENDATIONS

- Train observers to conduct interviews in a way that would lead to unbiased biosampling. This could be achieved by changing the order in which questions are asked (e.g., asking if fish could be biosampled before asking about clip incidence). If logistically possible, it would be interesting to use two different versions of the datasheet to test for an effect on clip-rate bias.
- More effort should be allocated to inspecting the catch of Coho Salmon for a missing adipose fin and to collecting the heads of these marked fish for submission to the Salmon Head Recovery Program, as these data will be useful in the future for fisheries management or stock assessment.
- Funding agencies should support researchers to prepare and publish the Area 3 and 4 recreational creel survey results from 2010-2014. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.

- A simple study should be conducted in order to provide an estimate of the proportion of angling vessels that are recorded along the standard flightpath. The study could involve two simultaneous aerial surveys, one following the standard flightpath and another doing a more in-depth survey that covers more ground.

ACKNOWLEDGEMENTS

We would like to thank the creel surveyors (Darren Bluebird, Brittany Fenwick, Sean Offutt, Renee Ryan, and Rebecca Smith) for collecting large amounts of data and biosamples from anglers; and Brittany Fenwick for data entry. Thanks to Pacific Salmon Commission, who funded this work through their Northern Fund competition. Thanks to Ivan Winther for acting as liaison between our team and DFO. Thanks also go to the administrative staff at LGL and NCSFNSS, whose work made it possible for this study to be conducted. We also would like to thank Karl English for providing useful comments on a draft of this report. Last but not least, we thank the many anglers, charter operators and fishing lodges who participated in the survey.

LITERATURE CITED

Benjamini, Y., and Y. Hochberg. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B* 57: 289-300.

Blakley, A. C., J. McKenzie, T. Joe, K. K. English, and W. B. Griffiths. 2003. Yukon River drainage basin harvest study, 2002. Report prepared for the Yukon River Basin Harvest Study Steering Committee, Council of Yukon First Nations, and Fisheries and Oceans, Canada, Whitehorse, YK.

Clark, W. G. 1992. Validation of the IPHC length-weight relationship for halibut. *International Pacific Halibut Commission Report of Assessment and Research Activities* 1991: 113-116.

English, K. K., G. F. Searing, and D. A. Nagtegaal. 2002. Review of Georgia Strait recreational creel survey, 1983-1999. *Canadian Technical Report of Fisheries and Aquatic Sciences* 2414: 81 p.

Goodman, L. 1960. On the exact variance of products. *Journal of the American Statistical Association* 55: 708-713.

Van Tongeren, V., and I. Winther. 2010. 2009 North Coast (Areas 3 & 4) creel survey statistics for salmon and groundfish. *Canadian Manuscript Report of Fisheries and Aquatic Sciences* 2907.

Zar, J. H. 1984. Biostatistical Analysis. Prentice-Hall Inc., Englewood Clifffs, NJ.

APPENDICES

APPENDIX 1: DATA FORMS.

2015 AREA 3/4 CREEL SURVEY
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE:______________________ FLIGHT#:_______

SURVEYOR:_______________________________

Note: only survey shaded subareas when conditions permit

Sub Area	RIGHT FRONT COUNT				TOTAL COUNT				FLIGHT TIME		SEA STATE (1-4)
	SPORT *(Fishing)*	SPORT *(Moving)*	COMM. *(Fishing)*	COMM. *(Moving)*	S *(F)*	S *(M)*	C *(F)*	C *(M)*	START	FINISH	
A											
C											
B											
C											
D											
E											
C											
F											
E											
F											
G											
L											
G											
J											
I											
K											
I											
H											
C											
A											

WEATHER / SEA STATE / COMMENTS:

SIGNATURE:_______________________________________

SHIFT SUMMARY SHEET

Landing Site: ____________ **Interviewer Name:** ____________

Interviewer Code: ____________ **Date:** ____________

Shift Code: ____________ **Shift Times:** ____________

Time Blocks	
From - To	TB
Before 7:00	1
7:00 - 7:59	2
8:00 - 8:59	3
9:00 - 9:59	4
10:00 - 10:59	5
11:00 - 11:59	6
12:00 - 12:59	7
13:00 - 13:59	8
14:00 - 14:59	9
15:00 - 15:59	10
16:00 - 16:59	11
17:00 - 17:59	12
18:00 - 18:59	13
19:00 - 19:59	14
20:00 - 20:59	15
21:00 - 21:59	16

* Fill in using numbers only.

Time Block	Fishing Conditions	Boats Missed	Not Fishing	Fishing
	Totals:			

Comments: ____________

Trailer Count: ________ **Time:** ________

Database Tally # :	(office use only)

2015 Area 3 and 4 Creel Survey

Site#__________ Obs# ________ Date __/__/__ DD/MM/YY Interview# __________

# of Anglers:_______ Guided?: Y N Depart. Time:______ Land Time:________

- [ ] Complete Form
- [ ] Fish Not Obs
- [ ] Adipose Not Checked
- [ ] Shellfish Only
- [ ] Refusal
- [ ] Incomplete Form

North Coast - Skeena First Nations Stewardship Society

Subarea:_________ Total Hours:___________

Species	Kept: Adipose Marked	Kept: Adipose Not Marked	Kept: Adipose Unk	Released Legal Size: Adipose Marked	Released Legal Size: Adipose Not Marked	Released Legal Size: Adipose Unk	Released Sublegal Size	Gear	# of Lines/ Traps

Target	Hours	Chart	Site

- [ ] 1 Before 0700
- [ ] 2 0700 0759
- [ ] 3 0800 0859
- [ ] 4 0900 0959
- [ ] 5 1000 1059
- [ ] 6 1100 1159
- [ ] 7 1200 1259
- [ ] 8 1300 1359
- [ ] 9 1400 1459
- [ ] 10 1500 1559
- [ ] 11 1600 1659
- [ ] 12 1700 1759
- [ ] 13 1800 1859
- [ ] 14 1900 1959
- [ ] 15 2000 2059
- [ ] 16 After 2100

Subarea:_________ Total Hours:___________

Species	Kept: Adipose Marked	Kept: Adipose Not Marked	Kept: Adipose Unk	Released Legal Size: Adipose Marked	Released Legal Size: Adipose Not Marked	Released Legal Size: Adipose Unk	Released Sublegal Size	Gear	# of Lines/ Traps

Target	Hours	Chart	Site

- [ ] 1 Before 0700
- [ ] 2 0700 0759
- [ ] 3 0800 0859
- [ ] 4 0900 0959
- [ ] 5 1000 1059
- [ ] 6 1100 1159
- [ ] 7 1200 1259
- [ ] 8 1300 1359
- [ ] 9 1400 1459
- [ ] 10 1500 1559
- [ ] 11 1600 1659
- [ ] 12 1700 1759
- [ ] 13 1800 1859
- [ ] 14 1900 1959
- [ ] 15 2000 2059
- [ ] 16 After 2100

Species	Sub Area	Fork Length (mm)	Sex N/C=0 Male=1 Female=2 Unable=9	Adipose Clip N=0, Y=1, N/C=9	Head Tag #	Scale Book #	Fish #	Flesh Colour R=0, W=1, M=2
				Chinook and coho only		Chinook Only		

Weather Conditions:_______________________________________

Comments:_______________________________________

Signature:

APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER BOAT TRIP) BY MONTH, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon caught per boat trip in June, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Chinook Salmon												
Guided	2.90	4.50	4.26	0.24	0.42	0.42	3.98	3.98	4.00	2.67	4.63	2.63
Unguided	1.79	2.00	1.33	0.46	1.14	0.59	1.59	1.00	2.18	0.64	1.58	
Combined	2.25	3.21	2.29	0.39	1.14	0.63	1.60	1.44	2.82	1.25	2.48	1.40
Chum Salmon												
Guided	0.10		0.04				0.02	0.02	0.02	0.17		
Unguided	0.07								0.01			
Combined	0.08								0.01	0.05		
Coho Salmon												
Guided	1.80	3.07	2.59	0.41	0.37	0.37	2.21	2.21	2.26	1.17	0.38	2.25
Unguided	2.79	1.87		0.95	0.28	0.29	0.33	0.63	1.48	1.36	1.16	0.71
Combined	2.38	2.45	0.86	0.78	0.28	0.28	0.33	0.67	1.75	1.30	0.93	1.53
Pink Salmon												
Guided							0.02	0.02	0.02			0.38
Unguided	0.07								0.02			
Combined	0.04								0.02			0.20
Sockeye Salmon												
Guided	0.10		0.04									
Unguided									0.02			
Combined	0.04								0.01			

Average number of salmon caught per boat trip in July, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Chinook Salmon												
Guided	1.38	1.09	1.31	0.14	0.57	0.71	3.67	2.47	2.42	2.20	1.93	0.14
Unguided	0.81	0.74	1.25	0.16	0.35	0.38	0.75	0.77	0.79	0.61	1.24	0.13
Combined	1.05	0.93	1.08	0.15	0.42	0.43	0.90	0.20	1.44	0.85	1.65	0.14
Chum Salmon												
Guided	0.29		0.15					0.04	0.05		0.13	
Unguided	0.10				0.01		0.04	0.02	0.02		0.05	
Combined	0.18		0.08		0.01		0.03		0.03		0.10	
Coho Salmon												
Guided	6.96	5.09	6.40	0.45	2.45	4.47	8.00	7.54	7.59	5.60	6.70	2.14
Unguided	4.16	3.26	4.25	0.58	1.98	1.57	2.46	3.38	4.05	3.71	3.00	0.50
Combined	5.38	4.24	5.33	0.54	2.08	2.01	2.75		5.45	4.00	5.18	1.69
Pink Salmon												
Guided	0.79	0.14	0.58		0.14	0.24	1.17	0.28	0.23	1.00	0.63	
Unguided	0.52	0.21	1.88		0.80	0.52	0.96	0.63	0.43	0.29	0.33	0.13
Combined	0.64	0.17	1.92		0.80	0.47	0.97		0.35	0.39	0.51	0.03
Sockeye Salmon												
Guided					0.05	0.12		0.02	0.02		0.03	
Unguided			0.13				0.01	0.00				
Combined			0.08			0.02	0.01		0.01		0.02	

Average number of salmon caught per boat trip in August, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Chinook Salmon												
Guided	0.81	0.75	0.83	0.43	0.22		0.87	0.87	0.88	0.37	0.59	0.04
Unguided	1.00	0.33	0.54	0.06	0.09	0.20	0.40	0.57	0.32	0.13	0.68	0.43
Combined	0.85	0.57	0.63	0.17	0.08	0.18	0.44	0.44	0.55	0.21	0.63	0.13
Chum Salmon												
Guided			0.02				0.01	0.01	0.01			
Unguided			0.08	0.03		0.02			0.01			
Combined			0.06	0.02		0.02			0.01			
Coho Salmon												
Guided	6.57	2.44	5.28	4.43	3.00	1.70	6.49	6.49	6.73	3.58	5.28	1.42
Unguided	3.40	3.25	4.31	1.03	0.71	1.22	2.00	2.29	2.78	1.75	1.79	0.43
Combined	5.96	2.79	5.50	2.04	0.71	1.27	2.00	2.44	4.40	1.89	3.98	1.19
Pink Salmon												
Guided	0.24	0.19	0.17		0.22	0.60	0.31	0.31	0.31	0.10	0.19	
Unguided		0.50	0.77	0.03	0.16	0.25	0.52		0.34	0.19	0.68	
Combined	0.19	0.32	0.63	0.02	0.15	0.29	0.50	0.11	0.33	0.16	0.37	
Sockeye Salmon												
Guided							0.01	0.01	0.01	0.02	0.03	
Unguided						0.02			0.02	0.06		
Combined						0.02			0.02	0.05	0.02	

Average number of groundfish caught per boat trip in June, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Cabezon												
Guided							0.02	0.02	0.02			
Unguided												
Combined									0.01			
Dogfish												
Guided												
Unguided											0.05	
Combined											0.04	
Greenling												
Guided												
Unguided					0.01							
Combined					0.01							
Lingcod												
Guided	0.30	0.57	0.41	0.06	0.05	0.05	0.79	0.79	0.82	0.33	0.50	1.63
Unguided	0.43	0.07		0.14	0.03	0.09	0.21	0.13	0.24		0.42	0.29
Combined	0.38	0.31		0.11	0.03	0.09	0.20	0.11	0.44	0.10	0.44	1.00
Pacific Halibut												
Guided	1.80	1.21	1.59	0.47	0.63	0.63	2.24	2.24	2.25	3.33	1.88	1.50
Unguided	1.21	0.40		0.49	0.45	0.76	0.56	1.50	1.24	0.79	1.79	2.57
Combined	1.46	0.79	0.57	0.48	0.45	0.79	0.55	1.78	1.59	1.55	1.81	2.00
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided							0.02	0.02	0.02			
Unguided									0.01			
Combined									0.01			
Starry Flounder												
Guided												
Unguided												
Combined												

Average number of groundfish caught per boat trip in July, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Cabezon												
Guided												
Unguided							0.01	0.01	0.01			
Combined							0.01		0.00			
Dogfish												
Guided												
Unguided												
Combined												
Greenling												
Guided								0.02	0.02			
Unguided	0.06							0.01	0.01			
Combined	0.04								0.01			
Lingcod												
Guided	0.38	0.82	0.52		0.02	0.06	0.33	0.34	0.34	0.60	0.03	0.71
Unguided	0.06	0.21			0.02	0.06	0.07	0.13	0.15	0.39	0.24	1.13
Combined	0.20	0.54			0.02	0.06	0.09	0.80	0.23	0.42	0.12	0.83
Pacific Halibut												
Guided	2.13	2.00	1.87	0.18	1.26	2.59		1.70	1.81	3.40	2.17	2.62
Unguided	0.45	0.47		0.02	0.44	0.51	0.56	0.75	0.87	1.39	1.19	2.13
Combined	1.18	1.29	0.17	0.08	0.47	0.82	0.53	0.60	1.24	1.70	1.76	2.48
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided					0.01							
Combined					0.01							
Starry Flounder												
Guided												
Unguided												
Combined												

Average number of groundfish caught per boat trip in August, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Cabezon												
Guided							0.01	0.01	0.01	0.05		0.13
Unguided									0.01			
Combined									0.01			0.10
Dogfish												
Guided							0.01	0.01	0.01			
Unguided												
Combined									0.00			
Greenling												
Guided												
Unguided			0.15									
Combined			0.13									
Lingcod												
Guided	0.14	1.00	0.50		0.15	0.40	0.27	0.27	0.29	0.15		0.38
Unguided	0.40	0.42	0.62		0.01	0.04	0.18	0.14	0.23	1.00	0.26	0.71
Combined	0.19	0.75	0.50		0.01	0.08	0.17	0.11	0.25	0.84	0.10	0.45
Pacific Halibut												
Guided	1.48	3.06	2.02	0.57	1.11	2.00	1.91	1.91	1.89	2.80	2.44	3.13
Unguided	2.20	1.17	1.62	0.24	0.71	0.87	0.78	1.57	1.22	1.31	1.05	2.00
Combined	1.62	2.25	1.56	0.34	0.71	0.99	0.81	2.11	1.50	1.74	1.92	2.87
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided					0.07	0.20						
Unguided				0.06								
Combined				0.04		0.02						
Starry Flounder												
Guided												
Unguided												
Combined												

Average number of rockfish caught per boat trip in June, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Black Rockfish												
Guided				0.29	0.26	0.26	0.17	0.17	0.18	0.83		
Unguided					0.10	0.09			0.10			
Combined				0.09	0.10	0.09			0.13	0.25		
Canary Rockfish												
Guided							0.02	0.02	0.02			
Unguided												
Combined									0.01			
China Rockfish												
Guided	0.20		0.07				0.03	0.03	0.03			
Unguided		0.07			0.04				0.01			
Combined	0.08	0.03			0.04				0.02			
Copper Rockfish												
Guided							0.06	0.06	0.07	0.17		
Unguided		0.13			0.03		0.08		0.03			
Combined		0.07			0.03		0.08		0.04	0.05		
Quillback Rockfish												
Guided							0.52	0.52	0.54	0.50		0.13
Unguided		0.20		0.14	0.03	0.13	0.23		0.22			1.14
Combined		0.10		0.09	0.03	0.13	0.23		0.33	0.15		0.60
Redstripe Rockfish												
Guided												
Unguided	0.07				0.01			0.13				
Combined	0.04				0.01			0.11				
Tiger Rockfish												
Guided							0.02	0.02	0.02			0.25
Unguided									0.03			
Combined									0.02			0.13
Yelloweye Rockfish												
Guided	0.20		0.11				0.16	0.16	0.16			0.38
Unguided		0.07				0.03	0.03		0.11		0.05	0.29
Combined	0.08	0.03	0.14			0.03	0.03		0.13		0.04	0.33
Yellowtail Rockfish												
Guided											0.25	
Unguided		0.13			0.01		0.03		0.02			
Combined		0.07			0.01		0.03		0.01		0.07	

Average number of rockfish caught per boat trip in July, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Black Rockfish												
Guided										0.40		
Unguided					0.01	0.19	0.02	0.02	0.02		0.05	0.38
Combined					0.01	0.16	0.02		0.01	0.06	0.02	0.10
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided	0.08		0.04					0.03	0.04	0.20		
Unguided							0.03	0.01		0.11		
Combined	0.04						0.03		0.01	0.12		
Copper Rockfish												
Guided												
Unguided					0.04	0.01		0.02	0.04			
Combined					0.04	0.01			0.02			
Quillback Rockfish												
Guided	0.04	0.23	0.12		0.02	0.06		0.17	0.18	0.40	0.07	0.48
Unguided	0.06				0.04	0.09	0.15	0.11	0.08	0.18	0.14	0.13
Combined	0.05	0.12			0.04	0.09	0.14		0.12	0.21	0.10	0.38
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided								0.01	0.01			0.05
Unguided							0.02	0.02	0.02			
Combined							0.02		0.01			0.03
Yelloweye Rockfish												
Guided	0.04	0.41	0.19					0.09	0.10			0.38
Unguided	0.06						0.06	0.06	0.05	0.11	0.05	2.00
Combined	0.05	0.22					0.06		0.07	0.09	0.02	0.83
Yellowtail Rockfish												
Guided		0.18	0.08									
Unguided						0.03		0.01	0.02			
Combined		0.10				0.03			0.01			

Average number of rockfish caught per boat trip in August, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Black Rockfish												
Guided		0.19	0.09				0.11	0.11	0.12	0.02		0.04
Unguided					0.03		0.07	1.57	0.15	0.88		
Combined		0.11			0.03		0.06	1.22	0.13	0.74		0.03
Canary Rockfish												
Guided		0.06	0.02				0.03	0.03	0.01	0.02		0.04
Unguided							0.02					
Combined		0.04					0.05		0.00			0.03
China Rockfish												
Guided							0.05	0.05	0.05	0.02		0.04
Unguided							0.02		0.03	0.25		
Combined							0.02		0.04	0.21		0.03
Copper Rockfish												
Guided												
Unguided					0.01		0.05					
Combined					0.01		0.05					
Quillback Rockfish												
Guided	0.38	0.56	0.41		0.07	0.20	0.22	0.22	0.23	0.32	0.03	0.75
Unguided		0.83	0.23	0.03	0.10	0.06	0.37		0.17	0.75	0.16	0.43
Combined	0.31	0.68	0.19	0.02	0.10	0.08	0.34		0.20	0.63	0.08	0.68
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided		0.13	0.04		0.04	0.10	0.02	0.02	0.02			
Unguided									0.02			0.43
Combined		0.07				0.01			0.02			0.10
Yelloweye Rockfish												
Guided		0.63	0.30				0.17	0.17	0.18	0.19	0.03	0.42
Unguided		0.17	0.31		0.10		0.08		0.05			0.71
Combined		0.43	0.25		0.10		0.08		0.10		0.02	0.48
Yellowtail Rockfish												
Guided							0.02	0.02	0.02			
Unguided					0.04				0.01			0.43
Combined					0.04				0.01			0.10

APPENDIX 3: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA AND SPECIES.

Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	820.5	1647.2	306.0	17.3	290.9	221.4	299.9	58.6	1708.7	89.4	372.2	205.7	6,038
Lodge Report	305.6	613.5	114.0	0	0	0	0	0	0	0	0	0	1,033
JUNE RETAINED	1126.0	2260.7	420.0	17.3	290.9	221.4	299.9	58.6	1708.7	89.4	372.2	205.7	7,071
SE	143.4	272.6	124.7	5.5	32.2	40.5	30.4	18.2	98.4	28.2	83.1	82.5	372
July													
Creel Estimate	354.2	360.9	149.6	22.8	159.5	245.9	291.5	13.9	1232.6	112.8	182.5	22.3	3,148
Lodge Report	355.5	362.3	150.1	0	0	0	0	0	0	0	0	0	868
JULY RETAINED	709.7	723.3	299.7	22.8	159.5	245.9	291.5	13.9	1232.6	112.8	182.5	22.3	4,016
SE	46.5	57.5	46.8	9.0	25.2	44.1	29.9	12.4	73.7	26.4	25.2	0	135
August													
Creel Estimate	378.8	123.5	91.9	2.7	20.9	100.1	112.7	4.9	519.6	38.2	51.8	22.6	1,468
Lodge Report	128.1	41.8	31.1	0	0	0	0	0	0	0	0	0	201
AUGUST RETAINED	506.9	165.3	123.0	2.7	20.9	100.1	112.7	4.9	519.6	38.2	51.8	22.6	1,669
SE	82.1	25.5	23.1	1.2	10.9	31.4	23.5	4.9	46.3	15.9	8.0	0	110
TOTAL													
Creel Estimate	1,553	2,132	547	43	471	567	704	77	3,461	240	607	251	10,654
Lodge Report	789	1,018	295	0	0	0	0	0	0	0	0	0	2,102
TOTAL RETAINED	2,343	3,149	843	43	471	567	704	77	3,461	240	607	251	12,756
SE	172	280	135	11	42	68	49	23	131	42	87	82	411

Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	15.2	495.9	38.3	4.1	61.8	189.8	103.1	0	439.4	3.6	33.3	9.8	1,394
Lodge Report	3.9	126.4	9.7	0	0	0	0	0	0	0	0	0	140
JUNE RELEASED	19.1	622.3	48.0	4.1	61.8	189.8	103.1	0	439.4	3.6	33.3	9.8	1,534
SE	12.4	190.4	28.5	2.2	14.5	77.3	34.9	0	65.2	2.7	17.2	7.5	222
July													
Creel Estimate	61.1	66.5	23.0	2.3	111.0	5.1	42.0	0	107.0	16.1	32.6	0	467
Lodge Report	51.5	56.1	19.4	0	0	0	0	0	0	0	0	0	127
JULY RELEASED	112.6	122.6	42.4	2.3	111.0	5.1	42.0	0	107.0	16.1	32.6	0	594
SE	27.7	33.8	11.2	1.9	40.1	3.9	12.8	0	28.6	8.6	15.2	0	70
August													
Creel Estimate	0	15.4	0	0	0	5.9	0	1.2	33.4	9.6	3.2	0	69
Lodge Report	0	34.0	0	0	0	0	0	0	0	0	0	0	34
AUGUST RELEASED	0	49.4	0	0	0	5.9	0	1.2	33.4	9.6	3.2	0	103
SE	0	11.0	0	0	0	4.2	0	1.2	12.8	6.8	1.6	0	19
TOTAL													
Creel Estimate	76	578	61	6	173	201	145	1	580	29	69	10	1,930
Lodge Report	55	216	29	0	0	0	0	0	0	0	0	0	301
TOTAL RELEASED	132	794	90	6	173	201	145	1	580	29	69	10	2,231
SE	30	194	31	3	43	78	37	1	72	11	23	7	234

Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2015.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
June													
Creel Estimate	866.0	1257.6	114.8	34.7	72.1	99.4	60.9	27.1	1063.6	93.0	138.9	225.3	4,053
Lodge Report	148.2	215.2	19.6	0	0	0	0	0	0	0	0	0	383
JUNE RETAINED	1014.2	1472.7	134.4	34.7	72.1	99.4	60.9	27.1	1063.6	93.0	138.9	225.3	4,436
SE	237.9	254.5	85.6	14.5	14.9	34.8	21.5	8.4	87.7	34.1	41.0	70.0	383
July													
Creel Estimate	1807.5	1652.8	736.3	79.7	794.0	1152.7	885.6	0	4670.4	531.7	573.6	273.2	13,158
Lodge Report	836.9	765.2	340.9	0	0	0	0	0	0	0	0	0	1,943
JULY RETAINED	2644.4	2418.0	1077.3	79.7	794.0	1152.7	885.6	0	4670.4	531.7	573.6	273.2	15,101
SE	147.6	216.5	109.1	28.0	78.5	143.3	74.4	0	174.1	69.8	52.7	0	389
August													
Creel Estimate	2668.9	602.2	808.5	31.9	177.8	695.1	515.2	27.1	4134.3	344.1	328.5	209.2	10,543
Lodge Report	1353.6	305.4	410.0	0	0	0	0	0	0	0	0	0	2,069
AUGUST RETAINED	4022.4	907.6	1218.5	31.9	177.8	695.1	515.2	27.1	4134.3	344.1	328.5	209.2	12,612
SE	340.6	102.1	127.6	10.1	33.5	93.2	61.5	15.3	205.0	70.8	38.9	0	453
TOTAL													
Creel Estimate	5,342	3,512	1,660	146	1,044	1,947	1,462	54	9,868	969	1,041	708	27,753
Lodge Report	2,339	1,286	771	0	0	0	0	0	0	0	0	0	4,395
TOTAL RETAINED	7,681	4,798	2,430	146	1,044	1,947	1,462	54	9,868	969	1,041	708	32,148
SE	441	349	188	33	87	174	99	17	283	105	77	70	709

Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2015.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
June													
Creel Estimate	76.0	35.4	0	0	33.5	4.5	42.2	9.0	94.2	7.2	0	0	302
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	76.0	35.4	0	0	33.5	4.5	42.2	9.0	94.2	7.2	0	0	302
SE	30.9	27.8	0	0	15.9	3.3	22.0	6.5	40.2	5.3	0	0	65
July													
Creel Estimate	836.6	417.9	46.0	18.2	260.0	71.7	238.2	417.8	394.6	44.3	67.4	0	2,813
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	836.6	417.9	46.0	18.2	260.0	71.7	238.2	417.8	394.6	44.3	67.4	0	2,813
SE	226.4	158.0	25.6	7.4	67.2	27.5	55.2	371.1	76.6	18.7	38.9	0	480
August													
Creel Estimate	34.4	123.5	0	0.7	0	35.3	20.1	0	181.8	19.1	1.6	5.7	422
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	34.4	123.5	0	0.7	0	35.3	20.1	0	181.8	19.1	1.6	5.7	422
SE	25.6	47.7	0	0.7	0	15.5	10.8	0	44.3	13.6	1.1	0	74
TOTAL													
Creel Estimate	947	577	46	19	294	112	301	427	671	71	69	6	3,537
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	947	577	46	19	294	112	301	427	671	71	69	6	3,537
SE	230	167	26	7	69	32	60	371	97	24	39	0	490

Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	15.2	0	0	0	0	0	0	0	10.5	0	0	29.4	55
Lodge Report	4.0	0	0	0	0	0	0	0	0	0	0	0	4
JUNE RETAINED	19.2	0	0	0	0	0	0	0	10.5	0	0	29.4	59
SE	12.4	0	0	0	0	0	0	0	4.3	0	0	12.0	18
July													
Creel Estimate	213.7	66.5	264.6	0	305.1	271.5	313.9	0	299.7	52.4	56.5	5.6	1,850
Lodge Report	2.4	0.7	2.9	0	0	0	0	0	0	0	0	0	6
JULY RETAINED	216.1	67.2	267.5	0	305.1	271.5	313.9	0	299.7	52.4	56.5	5.6	1,856
SE	62.2	21.7	54.3	0	56.2	51.5	52.7	0	38.1	17.8	18.7	0	134
August													
Creel Estimate	86.1	69.5	91.9	0.3	38.3	159.0	128.8	1.2	308.0	28.7	30.7	0	943
Lodge Report	0.3	0.3	0.4	0	0	0	0	0	0	0	0	0	1
AUGUST RETAINED	86.4	69.8	92.2	0.3	38.3	159.0	128.8	1.2	308.0	28.7	30.7	0	944
SE	37.1	23.8	35.5	0.3	10.2	31.8	24.0	1.2	60.5	14.9	8.7	0	94
TOTAL													
Creel Estimate	315	136	356	0	343	431	443	1	618	81	87	35	2,847
Lodge Report	7	1	3	0	0	0	0	0	0	0	0	0	11
TOTAL RETAINED	322	137	360	0	343	431	443	1	618	81	87	35	2,858
SE	73	32	65	0	57	60	58	1	72	23	21	12	165

Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	3.5	0	0	0	3
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	3.5	0	0	0	3
SE	0	0	0	0	0	0	0	0	2.5	0	0	0	3
July													
Creel Estimate	164.9	721.9	759.4	0	374.5	517.5	266.2	0	978.7	177.2	60.8	0	4,021
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	164.9	721.9	759.4	0	374.5	517.5	266.2	0	978.7	177.2	60.8	0	4,021
SE	52.4	262.2	496.7	0	106.4	183.8	75.2	0	151.1	99.1	26.2	0	634
August													
Creel Estimate	292.7	239.3	45.9	3.3	125.5	541.9	309.9	34.5	842.4	181.6	106.8	0	2,724
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	292.7	239.3	45.9	3.3	125.5	541.9	309.9	34.5	842.4	181.6	106.8	0	2,724
SE	164.2	87.1	32.6	3.3	54.1	158.1	90.3	19.3	137.2	59.5	47.1	0	311
TOTAL													
Creel Estimate	458	961	805	3	500	1,059	576	35	1,825	359	168	0	6,749
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	458	961	805	3	500	1,059	576	35	1,825	359	168	0	6,749
SE	172	276	498	3	119	242	118	19	204	116	54	0	706

Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2015.

Month / Catch Type	Subarea 3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	30.4	0	0	0	0	0	0	0	7.0	3.6	0	0	41
Lodge Report	23.0	0	0	0	0	0	0	0	0	0	0	0	23
JUNE RETAINED	53.4	0	0	0	0	0	0	0	7.0	3.6	0	0	64
SE	17.1	0	0	0	0	0	0	0	3.5	2.7	0	0	18
July													
Creel Estimate	61.1	0	11.5	0	3.5	0	11.2	0	24.5	0	10.9	0	123
Lodge Report	20.2	0	3.8	0	0	0	0	0	0	0	0	0	24
JULY RETAINED	81.3	0	15.3	0	3.5	0	11.2	0	24.5	0	10.9	0	147
SE	27.0	0	8.3	0	2.6	0	5.1	0	7.7	0	4.0	0	30
August													
Creel Estimate	0	0	9.2	0.3	0	11.8	0	0	7.4	0	0	0	29
Lodge Report	0.0	0.0	10.0	0	0	0	0	0	0	0	0	0	10
AUGUST RETAINED	0	0	19.2	0.3	0	11.8	0	0	7.4	0	0	0	39
SE	0	0	6.5	0.3	0	8.4	0	0	3.8	0	0	0	11
TOTAL													
Creel Estimate	91	0	21	0	3	12	11	0	39	4	11	0	192
Lodge Report	43	0	14	0	0	0	0	0	0	0	0	0	57
TOTAL RETAINED	135	0	34	0	3	12	11	0	39	4	11	0	249
SE	32	0	11	0	3	8	5	0	9	3	4	0	37

Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2015.

Month / Catch Type	Subarea 3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	12.2	0	0	0	0	0	5.6	0	6.1	0	0	0	24
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	12.2	0	0	0	0	0	5.6	0	6.1	0	0	0	24
SE	6.1	0	0	0	0	0	4.2	0	3.2	0	0	0	8
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	12	0	0	0	0	0	6	0	6	0	0	0	24
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	12	0	0	0	0	0	6	0	6	0	0	0	24
SE	6	0	0	0	0	0	4	0	3	0	0	0	8

Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	15.2	0	0	0	0	0	0	0	7.0	0	0	0	22
Lodge Report	1.1	0	0	0	0	0	0	0	0	0	0	0	1
JUNE RETAINED	16.3	0	0	0	0	0	0	0	7.0	0	0	0	23
SE	12.4	0	0	0	0	0	0	0	3.5	0	0	0	13
July													
Creel Estimate	0	0	11.5	0	0	10.2	2.8	0	6.1	0	2.2	0	33
Lodge Report	0	0	0.9	0	0	0	0	0	0	0	0	0	1
JULY RETAINED	0	0	12.4	0	0	10.2	2.8	0	6.1	0	2.2	0	34
SE	0	0	8.3	0	0	7.7	2.1	0	3.2	0	1.5	0	12
August													
Creel Estimate	0	0	0	0	0	11.8	0	0	14.8	9.6	1.6	0	38
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	11.8	0	0	14.8	9.6	1.6	0	38
SE	0	0	0	0	0	5.9	0	0	8.5	6.8	1.1	0	12
TOTAL													
Creel Estimate	15	0	12	0	0	22	3	0	28	10	4	0	93
Lodge Report	1	0	1	0	0	0	0	0	0	0	0	0	2
TOTAL RETAINED	16	0	12	0	0	22	3	0	28	10	4	0	95
SE	12	0	8	0	0	10	2	0	10	7	2	0	22

Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	41.6	0	0	0	0	0	0	0	42
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	41.6	0	0	0	0	0	0	0	42
SE	0	0	0	0	30.9	0	0	0	0	0	0	0	31
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	0	0	0	0	42	0	0	0	0	0	0	0	42
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	42	0	0	0	0	0	0	0	42
SE	0	0	0	0	31	0	0	0	0	0	0	0	31

Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	531.8	407.4	76.5	21.5	115.8	280.1	103.1	72.2	966.0	110.8	272.2	293.9	3,251
Lodge Report	157.1	120.3	22.6	0	0	0	0	0	0	0	0	0	300
JUNE RETAINED	688.8	527.7	99.1	21.5	115.8	280.1	103.1	72.2	966.0	110.8	272.2	293.9	3,551
SE	118.7	113.9	57.1	7.9	15.6	35.9	22.8	17.3	58.5	30.4	53.1	41.2	204
July													
Creel Estimate	396.9	503.4	23.0	11.4	180.3	471.3	171.0	41.8	1064.4	225.5	195.5	401.4	3,686
Lodge Report	176.2	223.5	10.2	0	0	0	0	0	0	0	0	0	410
JULY RETAINED	573.2	727.0	33.2	11.4	180.3	471.3	171.0	41.8	1064.4	225.5	195.5	401.4	4,096
SE	66.0	78.0	16.7	7.8	31.4	56.6	25.7	37.1	63.2	37.7	19.5	0	151
August													
Creel Estimate	723.2	486.4	229.7	5.3	177.8	541.9	209.3	23.4	1406.5	315.5	158.6	503.1	4,781
Lodge Report	211.0	141.9	67.0	0	0	0	0	0	0	0	0	0	420
AUGUST RETAINED	934.2	628.3	296.7	5.3	177.8	541.9	209.3	23.4	1406.5	315.5	158.6	503.1	5,201
SE	133.6	60.7	45.7	2.4	26.7	55.1	28.9	10.0	78.9	64.2	15.1	0	197
TOTAL													
Creel Estimate	1,652	1,397	329	38	474	1,293	483	137	3,437	652	626	1,198	11,718
Lodge Report	544	486	100	0	0	0	0	0	0	0	0	0	1,130
TOTAL RETAINED	2,196	1,883	429	38	474	1,293	483	137	3,437	652	626	1,198	12,848
SE	190	151	75	11	44	87	45	42	117	80	59	41	321

Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	212.7	106.3	0	11.6	23.2	117.5	4.7	36.1	477.8	60.8	88.9	431.0	1,570
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	212.7	106.3	0	11.6	23.2	117.5	4.7	36.1	477.8	60.8	88.9	431.0	1,570
SE	83.7	70.4	0	5.8	10.7	37.5	3.3	25.8	72.7	25.5	28.4	170.7	224
July													
Creel Estimate	262.6	256.5	0	0	48.5	128.1	30.8	83.6	406.8	80.6	63.0	256.5	1,617
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	262.6	256.5	0	0	48.5	128.1	30.8	83.6	406.8	80.6	63.0	256.5	1,617
SE	88.4	113.8	0	0	19.4	37.5	13.7	74.2	80.3	29.2	23.0	0	190
August													
Creel Estimate	86.1	262.5	18.4	1.0	17.4	64.8	68.4	3.7	348.9	172.1	63.1	180.9	1,287
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	86.1	262.5	18.4	1.0	17.4	64.8	68.4	3.7	348.9	172.1	63.1	180.9	1,287
SE	32.1	81.5	13.0	1.0	12.9	31.9	42.4	3.7	61.8	59.9	24.1	0	137
TOTAL													
Creel Estimate	561	625	18	13	89	310	104	123	1,233	313	215	868	4,475
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	561	625	18	13	89	310	104	123	1,233	313	215	868	4,475
SE	126	157	13	6	26	62	45	79	125	71	44	171	324

Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	136.7	159.4	0	5.0	7.7	31.6	37.5	4.5	268.5	7.2	66.7	146.9	872
Lodge Report	78.5	91.5	0	0	0	0	0	0	0	0	0	0	170
JUNE RETAINED	215.2	250.9	0	5.0	7.7	31.6	37.5	4.5	268.5	7.2	66.7	146.9	1,042
SE	50.0	69.6	0	2.8	5.5	14.2	16.6	3.2	42.7	3.7	30.9	82.6	132
July													
Creel Estimate	67.2	209.0	0	0	6.9	35.9	28.0	55.7	195.7	56.4	13.0	133.8	802
Lodge Report	36.2	112.8	0	0	0	0	0	0	0	0	0	0	149
JULY RETAINED	103.4	321.7	0	0	6.9	35.9	28.0	55.7	195.7	56.4	13.0	133.8	951
SE	31.1	58.9	0	0	3.6	14.8	9.2	49.5	32.0	21.4	6.5	0	93
August													
Creel Estimate	86.1	162.1	73.5	0	3.5	41.2	44.3	1.2	237.5	153.0	8.1	79.1	890
Lodge Report	45.2	85.2	38.6	0	0	0	0	0	0	0	0	0	169
AUGUST RETAINED	131.3	247.3	112.1	0	3.5	41.2	44.3	1.2	237.5	153.0	8.1	79.1	1,059
SE	37.1	55.7	36.9	0	2.6	18.1	17.5	1.2	35.2	47.8	3.4	0	100
TOTAL													
Creel Estimate	290	531	73	5	18	109	110	61	702	216	88	360	2,563
Lodge Report	160	289	39	0	0	0	0	0	0	0	0	0	488
TOTAL RETAINED	450	820	112	5	18	109	110	61	702	216	88	360	3,051
SE	70	107	37	3	7	27	26	50	64	53	32	83	190

Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	15.2	0	0	0.8	0	0	0	0	7.0	0	0	0	23
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	15.2	0	0	0.8	0	0	0	0	7.0	0	0	0	23
SE	12.4	0	0	0.8	0	0	0	0	5.0	0	0	0	13
July													
Creel Estimate	0	0	0	2.3	6.9	0	0	0	0	0	0	0	9
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	2.3	6.9	0	0	0	0	0	0	0	9
SE	0	0	0	1.9	3.6	0	0	0	0	0	0	0	4
August													
Creel Estimate	0	15.4	0	0	0	11.8	16.1	0	3.7	0	0	0	47
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	15.4	0	0	0	11.8	16.1	0	3.7	0	0	0	47
SE	0	11.0	0	0	0	5.9	9.5	0	2.7	0	0	0	16
TOTAL													
Creel Estimate	15	15	0	3	7	12	16	0	11	0	0	0	79
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	15	15	0	3	7	12	16	0	11	0	0	0	79
SE	12	11	0	2	4	6	9	0	6	0	0	0	21

Estimated numbers of other groundfish* retained by lodge and non-lodge anglers, by month, 2015.

Month / Catch Type	Subarea 3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	15.2	0	0	0	7.7	18.1	4.7	0	24.4	0	11.1	0	81
Lodge Report	58.0	0	0	0	0	0	0	0	0	0	0	0	58
JUNE RETAINED	73.2	0	0	0	7.7	18.1	4.7	0	24.4	0	11.1	0	139
SE	12.4	0	0	0	3.1	10.4	3.3	0	7.5	0	6.7	0	20
July													
Creel Estimate	30.5	0	23.0	0	10.4	10.2	22.4	0	18.4	0	2.2	0	117
Lodge Report	18.2	0	13.8	0	0	0	0	0	0	0	0	0	32
JULY RETAINED	48.8	0	36.8	0	10.4	10.2	22.4	0	18.4	0	2.2	0	149
SE	15.6	0	16.7	0	4.4	5.4	7.8	0	6.4	0	1.5	0	26
August													
Creel Estimate	0	0	45.9	0.7	0	17.7	16.1	0	40.8	0	6.5	28.3	156
Lodge Report	0.0	0.0	48.0	0	0	0	0	0	0	0	0	0	48
AUGUST RETAINED	0	0	93.9	0.7	0	17.7	16.1	0	40.8	0	6.5	28.3	204
SE	0	0	23.5	0.7	0	9.4	7.3	0	14.4	0	3.6	0	30
TOTAL													
Creel Estimate	46	0	69	1	18	46	43	0	84	0	20	28	354
Lodge Report	76	0	62	0	0	0	0	0	0	0	0	0	138
TOTAL RETAINED	122	0	131	1	18	46	43	0	84	0	20	28	492
SE	20	0	29	1	5	15	11	0	17	0	8	0	44

* includes Cabezon, Dogfish, Starry Flounder, Greenling, Ratfish, Rock Sole, and unknown groundfish

Estimated numbers of other groundfish* released by lodge and non-lodge anglers, by month, 2015.

Month / Catch Type	Subarea 3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	30.4	0	0	0	0	4.5	14.1	0	0	0	0	0	49
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	30.4	0	0	0	0	4.5	14.1	0	0	0	0	0	49
SE	24.8	0	0	0	0	3.3	7.4	0	0	0	0	0	26
July													
Creel Estimate	0	0	0	0	0	20.5	0	0	3.1	0	0	0	24
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	20.5	0	0	3.1	0	0	0	24
SE	0	0	0	0	0	12.2	0	0	2.3	0	0	0	12
August													
Creel Estimate	0	0	0	0	7.0	11.8	8.1	0	3.7	9.6	1.6	5.7	47
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	7.0	11.8	8.1	0	3.7	9.6	1.6	5.7	47
SE	0	0	0	0	5.2	8.4	4.2	0	2.7	6.8	1.1	0	13
TOTAL													
Creel Estimate	30	0	0	0	7	37	22	0	7	10	2	6	120
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	30	0	0	0	7	37	22	0	7	10	2	6	120
SE	25	0	0	0	5	15	9	0	4	7	1	0	32

* includes Cabezon, Dogfish, Starry Flounder, Greenling, Ratfish, Rock Sole, and unknown groundfish

Estimated numbers of rockfish* retained by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	76.0	159.4	19.1	18.2	66.9	99.4	107.8	4.5	610.3	32.2	27.8	215.5	1,437
Lodge Report	58.5	122.8	14.7	0	0	0	0	0	0	0	0	0	196
JUNE RETAINED	134.5	282.2	33.9	18.2	66.9	99.4	107.8	4.5	610.3	32.2	27.8	215.5	1,633
SE	32.6	55.7	14.3	10.3	15.4	26.1	24.7	3.2	72.5	15.8	14.2	72.7	130
July													
Creel Estimate	158.8	247.0	0	0	62.4	199.8	213.0	0	532.2	120.8	36.9	312.2	1,883
Lodge Report	91.6	142.4	0	0	0	0	0	0	0	0	0	0	234
JULY RETAINED	250.3	389.4	0	0	62.4	199.8	213.0	0	532.2	120.8	36.9	312.2	2,117
SE	63.6	67.8	0	0	24.7	74.9	42.6	0	86.0	27.4	12.5	0	158
August													
Creel Estimate	172.2	401.4	82.7	1.7	90.6	94.2	265.7	13.6	664.3	305.9	38.8	401.4	2,532
Lodge Report	112.5	262.4	54.0	0	0	0	0	0	0	0	0	0	429
AUGUST RETAINED	284.7	663.9	136.7	1.7	90.6	94.2	265.7	13.6	664.3	305.9	38.8	401.4	2,961
SE	84.1	78.9	35.1	1.4	20.5	20.9	52.4	13.6	84.1	69.8	13.8	0	175
TOTAL													
Creel Estimate	407	808	102	20	220	393	586	18	1,807	459	104	929	5,853
Lodge Report	263	528	69	0	0	0	0	0	0	0	0	0	859
TOTAL RETAINED	670	1,335	171	20	220	393	586	18	1,807	459	104	929	6,712
SE	110	118	38	10	36	82	72	14	140	77	23	73	269

* includes Black, Canary, China, Copper, Quillback, Redstripe, Tiger, Yelloweye, Yellowtail and unknown rockfish

Estimated numbers of rockfish* released by lodge and non-lodge anglers, by month, 2015.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	15.2	0	0	0	2.6	31.6	28.1	0	38.4	0	0	0	116
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	15.2	0	0	0	2.6	31.6	28.1	0	38.4	0	0	0	116
SE	12.4	0	0	0	1.8	17.8	14.1	0	12.0	0	0	0	29
July													
Creel Estimate	30.5	0	0	0	24.3	25.6	16.8	0	21.4	12.1	0	5.6	136
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	30.5	0	0	0	24.3	25.6	16.8	0	21.4	12.1	0	5.6	136
SE	21.6	0	0	0	9.9	13.9	12.6	0	8.7	6.9	0	0	32
August													
Creel Estimate	0	0	0	0	0	5.9	20.1	0	40.8	76.5	3.2	17.0	164
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	5.9	20.1	0	40.8	76.5	3.2	17.0	164
SE	0	0	0	0	0	4.2	10.9	0	14.0	42.4	2.3	0	46
TOTAL													
Creel Estimate	46	0	0	0	27	63	65	0	101	89	3	23	416
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	46	0	0	0	27	63	65	0	101	89	3	23	416
SE	25	0	0	0	10	23	22	0	20	43	2	0	63

* includes Black, Canary, China, Copper, Quillback, Redstripe, Tiger, Yelloweye, Yellowtail and unknown rockfish

APPENDIX 4: HARVEST DIFFERENCES AMONG MONTHS.

Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species. Subareas treated as replicates.

	Kruskal-Wallis Statisitics			*Median Monthly Harvest*		
Species	χ^2	*df*	*P*	*June*	*July*	*August*
Chinook Salmon	12.0	2	0.002	139.9	86.2	32.2
Coho Salmon	10.0	2	0.007	50.3	341.7	186.3
Pink Salmon	28.6	2	< 0.0001	0	62.8	27.1
Chum Salmon	5.3	2	0.07	0	0.6	0
Sockeye Salmon	2.9	2	0.24	0	0	0
Pacific Halibut	2.2	2	0.33	67.9	117.3	138.7
Lingcod	0.1	2	0.94	14.6	17.4	29.4
Cabezon	1.1	2	0.59	0	0	0
Dogfish	2.1	2	0.35	0	0	0
Greenling	1.1	2	0.57	0	0	0
Rock Sole	0.4	2	0.83	0	0	0
Black Rockfish	0.8	2	0.66	0	1.2	2.4
Canary Rockfish	12.6	2	0.002	0	0	0
China Rockfish	0.0	2	0.99	0	0	0
Copper Rockfish	4.6	2	0.10	0	0	0
Quillback Rockfish	6.1	2	0.048	3.9	9.5	25.6
Redstripe Rockfish	12.9	2	0.002	0	0	0
Tiger Rockfish	1.9	2	0.39	0	0	0
Yelloweye Rockfish	0.2	2	0.92	4.0	2.3	3.7
Yellowtail Rockfish	0.9	2	0.62	0	0	0